Random matrix theory and Dyson-Brownian-motion diagnostics for breast cancer coexpression networks: a Monte-Carlo-validated framework applied to TCGA-BRCA

Julio César Galindo López[1,*]
Enrique Hernández-Lemus[2]

[1] Facultad de Ciencias Empresariales, Universidad Panamericana, Ciudad de México
[2] Computational Systems Biology and Integrative Genomics Laboratory, Instituto Nacional de Medicina Genómica (INMEGEN)

*Correspondence: Julio César Galindo López, jcgalindo@up.edu.mx

September 2026

Abstract

Random matrix theory (RMT) offers a principled way to decide which correlations in a noisy, finite-sample gene-expression matrix reflect real biology and not sampling noise. We combine a classical RMT correlation threshold, a Monte Carlo permutation null, and a Dyson-Brownian-motion repulsion diagnostic into one pipeline and apply it to real breast cancer data from The Cancer Genome Atlas (TCGA-BRCA), comparing triple-negative (TNBC, $n = 119$) and luminal A (LumA, $n = 430$) cohorts. The RMT threshold separates the two cohorts, a gap confirmed by a permutation null; spiked eigenvalues beyond the Marchenko–Pastur bulk correspond to biologically coherent gene programs, not artifacts. Tracking eigenvalues as the sample grows reveals Dyson-type level repulsion, reproducible across 20 random orders in both cohorts, with a new corollary explaining which pairs repel least reliably. The same pipeline extends, for the first time, to two further data modalities: single-cell RNA-seq, recovering a real threshold in a regime bulk data cannot reach and separating true cell-type signal from technical artifacts; and Hi-C chromatin contact data, recovering a compartment structure independently validated against real GC content and gene density, replicated across three triple-negative tumors and a matched normal sample. Three negative results are reported, not dropped: a wavelet decomposition fails identically under three unrelated cell-ordering axes, ruling out ordering choice as the explanation; a classifier reaches near-ceiling accuracy regardless of feature-selection method, showing the benchmark cannot discriminate methods; and a trained classifier's own weight-matrix eigenvalues repel only during the pre-convergence training transient. All six research lines originally proposed for this program have now been run on real data at least once, with code and data fully reproducible. This validation-first approach, pairing every positive finding with an explicit statistical null and reporting negative results with equal rigor, offers a template for computational-biology pipelines built on random matrix theory.

Author summary

Breast cancer is not one disease: triple-negative and luminal A tumors differ enough that grouping them together can hide real biology. We ask a mathematical question about this difference: when we measure how strongly every pair of genes moves together across many patients, how do we tell a

real biological signal from statistical noise? We borrow a tool from physics, random matrix theory, originally developed to understand atomic-nucleus energy levels, that gives a principled threshold for this, plus a permutation test confirming the threshold is not an accident of the math. We apply this to real tumor data, extend it for the first time to single cells and to the three-dimensional folding of the genome, and check every positive finding against independent evidence. We are equally careful to report what did not work: a method for separating fast from slow biological change, a machine-learning benchmark that turned out too easy, and a training-dynamics diagnostic that only sees part of what it was built to see. All of it, code, data, successes and failures alike, is included so the next researcher can check, reuse, or correct any of it.
Keywords: random matrix theory; gene coexpression networks; triple-negative breast cancer; TCGA; Monte Carlo permutation test; Dyson Brownian motion; Marchenko–Pastur law

1 Introduction

Triple-negative breast cancer (TNBC) — tumors negative for estrogen receptor, progesterone receptor, and HER2 — lacks the targeted therapies available for hormone-receptor-positive or HER2-enriched disease, and remains transcriptionally and clinically heterogeneous [11]. Network-based approaches that treat gene coexpression as a graph, rather than testing genes one at a time, have become a standard tool for finding this heterogeneity's structure, and the Hernández-Lemus group has applied several such approaches directly to breast cancer: multilayer information-theoretic transcriptional networks [26], k-core structural analysis [7], and multi-omic subtype-specific interaction networks [27]. The group's own network-inference pipeline for breast cancer, publicly available as the CSB-IG `breast_cancer_networks` and `parallel-aracne` repositories, is built on ARACNE [25]: mutual information between every gene pair, pruned by the Data Processing Inequality, with a bootstrap procedure (the NIMEA pipeline, `CSB-IG/BioNetworkInference_ModularAnalysis`) used to set the mutual-information significance threshold and Infomap used for module detection — an information-theoretic route to the same underlying question this paper asks with a spectral one: given a noisy, finite-sample gene-expression matrix, which pairwise associations are real? The RMT threshold in Sec. 2.4 and the group's own bootstrap-calibrated mutual-information threshold are two different statistical routes to a threshold-selection problem neither one has an obviously superior claim to; running both on the same TNBC cohort and comparing the resulting hub genes and modules is a direct, concrete extension of this paper, using tooling the group already has in production rather than anything new.

A separate line of work asks a narrower but sharper question: given a correlation matrix built from noisy, finite-sample gene-expression data, which correlations reflect real biological structure and which are sampling noise? Random matrix theory answers this by comparing the empirical eigenvalue spectrum against the null spectrum expected from a random (structureless) matrix of the same size. Luo et al. [22] formalized this into a concrete recipe for gene coexpression networks: sweep a correlation threshold τ, and select the value at which the nearest-neighbor eigenvalue spacing distribution switches from Poisson (uncorrelated, structureless) to the Gaussian-orthogonal-ensemble (GOE) Wigner surmise (correlated, structured) — a transition driven by level repulsion, the same mechanism Wigner and Dyson identified in nuclear spectra and later in general Hermitian random matrices [8]. The method has since been used at genome scale [12] and specifically in cancer gene networks [17].

Two further questions motivate the present paper. First, is the selected threshold real, or an artifact of the asymptotic Marchenko–Pastur/Wigner-surmise null used to select it? A Monte Carlo permutation null — shuffling each gene's values across patients independently, destroying every real correlation while keeping each gene's own marginal distribution fixed, the same kind of null the group's own NIMEA pipeline uses to calibrate its mutual-information threshold (Sec. 2.5, below) — answers this directly rather than relying on asymptotics alone. Second, Aarts et al. [1] showed that the weight matrices of a neural network undergoing stochastic-gradient-descent training evolve according to Dyson's (1962) eigenvalue Brownian motion, with the same level-repulsion mechanism at work — suggesting that the repulsion diagnostic itself, not just the static threshold it eventually selects, carries information, and that it applies equally to a biological correlation matrix and to the training dynamics of a classifier built on top of it.

This paper makes three contributions. First, it states, as a single explicit framework, how a static RMT threshold, a Monte Carlo permutation null, and a Dyson-repulsion diagnostic relate to one another — the same underlying repulsion mechanism observed two different ways, not two unrelated tools used side by side (Methods). Second, it runs that framework, together with a machine-learning recognition test, across three real, independent data modalities from the same disease — bulk RNA-seq (TCGA-BRCA, comparing TNBC against luminal A breast cancer), single-cell RNA-seq, and Hi-C chromatin contact data — not on synthetic data, and not as a proposal: this is, to our knowledge, the first application of a Dyson-repulsion diagnostic to real gene-expression data at any resolution, and the first time a single RMT pipeline has been run, unmodified beyond the one data-specific preprocessing step each modality requires, across bulk, single-cell, and 3D genomic data for the same cancer type. Third, it reports every negative result the same way it reports every positive one, without adjustment or omission: a wavelet multiscale method that fails identically under three unrelated ordering axes, a classification benchmark too easy to discriminate feature-selection methods, and a classifier training diagnostic that sees only part of what it was built to see. A validation-first pipeline is only as trustworthy as its negative results are honest; this paper treats that discipline as a result in itself, not as a shortfall to be hidden.

1.1 Mathematical background

We collect here, precisely, the four pieces of random matrix theory the rest of the paper uses, since each plays a distinct role and the distinctions matter for how the results should be read.

Definition 1 (Wigner ensemble and the GOE)**.** *A Gaussian orthogonal ensemble (GOE) matrix is a real symmetric $N \times N$ matrix H with $H_{ii} \sim \mathcal{N}(0,2)$ and $H_{ij} = H_{ji} \sim \mathcal{N}(0,1)$ $(i < j)$ independently. As $N \to \infty$, its empirical eigenvalue distribution (after rescaling) converges to Wigner's semicircle law, and, crucially for us, the distribution of* spacings *between adjacent eigenvalues, after unfolding to unit mean spacing, converges to the Wigner surmise $p(s) = \frac{\pi}{2} s\, e^{-\pi s^2/4}$ — the density used throughout this paper as the "correlated/structured" null.*

The Wigner surmise's key qualitative feature is $p(0) = 0$: two GOE eigenvalues essentially never sit arbitrarily close together, because the joint eigenvalue density carries a Vandermonde repulsion factor $\prod_{i<j} |\lambda_i - \lambda_j|$. This is the same repulsion term that appears, dynamically rather than statically, in Dyson's model below; the static NNSD test and the dynamic trajectory test in Sec. 2.9 are two different windows onto the same underlying mechanism, not two unrelated diagnostics. By contrast, a matrix with no such structure (e.g. a diagonal matrix of i.i.d. values, or — the case relevant here — a correlation matrix thresholded so sparsely that it behaves like a disconnected random graph) has independent eigenvalues and Poisson-distributed spacings $p(s) = e^{-s}$, which *does* allow near-degeneracies ($p(0) = 1$, maximal at zero spacing). The Luo et al. method [22] used throughout Sec. 2.4 operationalizes exactly this contrast: sparse/uninformative thresholds

look Poisson, and the threshold at which the spectrum first looks GOE-like is read as the point where real correlation structure first dominates chance.

Remark 1 (The $N \to \infty$ limit, free probability, and Hilbert spaces)**.** *The Wigner semicircle law referenced above is a statement about $N \times N$ GOE matrices as $N \to \infty$: the empirical spectral measure converges to a fixed limiting shape. Free probability theory [37] recasts this convergence operator-theoretically: a sequence of independent GOE (or Wigner) matrices converges, in the sense of joint moments, to a family of* free *self-adjoint elements in a von Neumann algebra equipped with a trace — a noncommutative probability space built on a Hilbert space, in exactly the sense operator algebras use the term. In that limit the semicircle law is not an analogy to the Gaussian but its literal free-probabilistic counterpart: it is the fixed point of free convolution the way the Gaussian is the fixed point of classical convolution (the free Central Limit Theorem), and "freeness" is the operator-algebraic substitute for classical independence that makes this work. None of this machinery is invoked computationally in this paper — every matrix here is finite ($p = 1500$ or 60), and every claim is checked directly on that finite matrix or by finite-n Monte Carlo, never by appeal to the $N \to \infty$ operator-algebraic limit — but it is the reason the semicircle/GOE machinery generalizes so far beyond Gaussian random matrices in the first place: free probability, not Gaussianity, is the real source of the universality being leaned on throughout Sec. 2.4 and Sec. 2.5. A standard modern reference for this whole circle of ideas, including the Dyson Brownian motion and Tracy–Widom material below, is Anderson et al. [3]. Ben Arous and Guionnet sharpened the convergence itself into a large deviations principle [5]: the empirical spectral measure of a GOE matrix does not merely converge to the semicircle law, it does so with an exponentially small probability of any other outcome, governed by a rate function built from Voiculescu's free (non-commutative) entropy — the same notion of entropy invoked, in the unrelated but same-named finite-matrix sense, by the spectral entropy diagnostic of Sec. 2.3. The two entropies are different objects (one a large-deviations rate functional on the space of limiting measures, the other a Shannon entropy of one finite matrix's own normalized spectrum) and this paper does not conflate them; they share a name, and a spectral origin, for a real reason worth recording rather than treating as coincidence.*

Remark 2 (Four-moment universality: why finite, non-Gaussian data is not a problem)**.** *The free-probability remark above addresses the $N \to \infty$ limit; a sharper, finite-N statement addresses the objection that TCGA-BRCA expression data is not Gaussian at all. Tao & Vu [33] proved a Four Moment Theorem: local eigenvalue statistics of a Wigner matrix (spacing distributions, correlation functions, and in particular the GOE/Wigner-surmise statistics used throughout Sec. 2.4) are determined, to leading order, only by the first four moments of the entry distribution, for both Hermitian and real symmetric ensembles — two matrices with matching mean, variance, skewness, and kurtosis have asymptotically indistinguishable local spectral statistics, regardless of whether either is Gaussian. Tao & Vu [34] extend this specifically to sample covariance/ correlation matrices, the exact object built in Sec. 6.1, establishing universality of local eigenvalue statistics under the same mild moment-matching condition. This is the precise, rather than merely qualitative, justification for comparing a real, non-Gaussian gene-expression correlation matrix's spacing statistics against the GOE Wigner surmise (Sec. 2.4) and the Marchenko–Pastur bulk (below): universality here is a theorem about matching moments, not an appeal to an idealization the data does not satisfy.*

Definition 2 (Marchenko–Pastur law)**.** *For a $p \times n$ data matrix with i.i.d. zero-mean, unit-variance entries and $p/n \to q \in (0, \infty)$, the eigenvalues of the sample correlation matrix converge to the Marchenko–Pastur distribution, supported on $[\lambda_-, \lambda_+] = [(1-\sqrt{q})^2, (1+\sqrt{q})^2]$ [24]. This is the null spectrum against which Sec. 2.2's spiked-eigenvalue count is taken: it says what the bulk of the spectrum would look like if none of the p genes were really correlated with each other, purely as an artifact of estimating a correlation matrix from a finite sample of n patients.*

An eigenvalue exceeding λ_+ is suggestive of real low-rank structure (a "spike"), not merely an artifact of q; the BBP transition [4] names the threshold, as a function of the true spike strength and q, above which a spike is detectable at all above the bulk. We use the count of observed spikes only descriptively (Table 2) and do not fit a formal spike-strength model in this paper; we also do not claim every eigenvalue above λ_+ is individually significant in a formal hypothesis-testing sense — finite-n edge fluctuations of the bulk itself follow a Tracy–Widom law [35] at the $n^{-2/3}$ scale, so a handful of eigenvalues only marginally above λ_+ could in principle be bulk fluctuations rather than true spikes. Checking this directly: of TNBC's 14 spiked eigenvalues, 7 exceed λ_+ by a factor of at least 2 (up to 7.6× for the largest) and are implausible as edge fluctuations, while the remaining 7 sit within a factor of 1.04–1.8 × λ_+ and are the ones this caveat applies to directly; LumA is similar in kind but better separated in practice (13 of 29 spikes exceed 2 × λ_+, up to 25× for the largest, reflecting its larger n and correspondingly less noisy bulk edge). Table 2 reports the top eigenvalue only; the full ratio-to-λ_+ list for every spike, in both cohorts, is in `results/spiked_eigenvalues.csv`.

Remark 3 (Painlevé II and why the edge is hard, not just noisy)**.** *The $n^{-2/3}$ edge-fluctuation scale just invoked is not merely a rate: Tracy & Widom [35] showed the limiting distribution of the (centered, rescaled) largest eigenvalue at that scale is governed by a particular (Hastings–McLeod) solution of the Painlevé II differential equation, $q''(x) = xq(x) + 2q(x)^3$, $q(x) \sim \mathrm{Ai}(x)$ as $x \to \infty$. The precise distribution depends on the ensemble's symmetry class (Dyson's β, Sec. 1.1 above): for GUE ($\beta = 2$), $F_2(s) = \exp\big(-\int_s^\infty (x-s)q(x)^2\,dx\big)$; for GOE ($\beta = 1$) — the class every correlation matrix in this paper belongs to, being real symmetric — the relation is more involved, $[F_1(s)]^2 = F_2(s)\exp\big(-\int_s^\infty q(x)\,dx\big)$ [36], not simply F_2 itself. Either way, the Tracy–Widom distributions are not Gaussian, not exponential, and have no elementary closed form; they are defined* through *a nonlinear ODE with no elementary solution. This paper does not fit that distribution directly (no verified numerical implementation of it was available), and instead checks a strictly weaker, fully elementary consequence of the same underlying scaling law: whether the standard deviation of a synthetic null's top eigenvalue shrinks like $n^{-2/3}$ as n grows at fixed aspect ratio q (Results, "Edge-fluctuation scaling"), a claim checkable without solving Painlevé II.*

Definition 3 (Spectral unfolding)**.** *Because the local density of eigenvalues varies across the spectrum even under a fixed null, raw spacings $\lambda_{k+1} - \lambda_k$ are not directly comparable to $p(s)$ above; unfolding replaces λ_k with $\hat{F}(\lambda_k)\cdot N$, where $\hat{F}$ is a smooth fit to the empirical eigenvalue CDF (here, a degree-7 polynomial, following standard RMT practice and 22), so that the unfolded spacings have mean 1 everywhere in the spectrum and can be compared to $p(s)$ directly. All NNSD comparisons in this paper (Sec. 2.4, Fig. 3) use unfolded spacings.*

Definition 4 (Dyson Brownian motion)**.** *Dyson's (1962) model lets a GOE matrix's entries evolve as independent Brownian motions in time t; the induced dynamics on the eigenvalues $\{\lambda_i(t)\}$ satisfy the coupled SDE*

$$d\lambda_i = dB_i + \sum_{j \neq i} \frac{dt}{\lambda_i - \lambda_j},$$

a Dyson Brownian motion [8]: the eigenvalues perform independent Brownian motion plus *a deterministic Coulomb-gas repulsion term that diverges as any two eigenvalues approach each other, mechanically preventing crossings (level repulsion in continuous time, the dynamical counterpart of the static Wigner surmise above). This process is unconfined: with no restoring drift pulling the λ_i back toward the origin, it has no stationary law of its own and its eigenvalues spread diffusively without bound as $t \to \infty$, exactly as a single Brownian motion does — a feature, not an omission,* *for how this paper uses it: "time" below stands for a steadily growing patient or cell sample, which never equilibrates either, so the absence of a stationary law is the correct match, not a gap to be patched. Aarts et al. [1] show this same SDE, empirically, governs the eigenvalues of a neural network's weight matrices under SGD training; Sec. 2.9 asks whether the same qualitative repulsion signature (a trajectory approaching, then visibly reopening, an adjacent trajectory rather than crossing it) is visible when "time" is instead sample size, applied directly to a real gene-correlation matrix rather than a trained model.*

Proposition 1 (A confined variant has the GOE density as its exact stationary law)**.** *The unconfined process of Definition 4 has no stationary law, so it cannot itself be the process the static Wigner-surmise test and the dynamic trajectory test share. What they do share is the confined Dyson SDE obtained by adding one linear (Ornstein–Uhlenbeck) restoring term and halving the repulsion coefficient to match,*

$$d\lambda_i = dB_i + \Big[\tfrac{1}{2}\sum_{j\neq i}\frac{1}{\lambda_i-\lambda_j} - \frac{\lambda_i}{4}\Big]dt, \qquad i = 1,\ldots,N,$$

a genuinely different SDE from Definition 4's (same divergent repulsion term in form, confined rather than unconfined, and rescaled to match the GOE normalization exactly), introduced here only because it is the process for which the following holds: the GOE joint eigenvalue density

$$p(\lambda_1,\ldots,\lambda_N) \;\propto\; \prod_{i<j}|\lambda_i-\lambda_j| \;\exp\Big(-\tfrac{1}{4}\sum_i \lambda_i^2\Big)$$

is its unique stationary distribution, and starting the SDE from any initial configuration and letting $t\to\infty$, the law of $(\lambda_1(t),\ldots,\lambda_N(t))$ converges to p.

Proof. For an SDE $d\lambda_i = dB_i + b_i\,dt$ with unit-variance independent driving noise, the Fokker–Planck generator has diffusion coefficient $\frac{1}{2}$, so a density p satisfying the stronger zero-current (detailed-balance) condition $b_i = \frac{1}{2}\partial_i \log p$ for every i is automatically a stationary solution; ergodicity (convergence from any initial law) then follows from standard results for uniformly elliptic diffusions with a strictly convex confining potential [3], which the added $-\lambda_i/4$ term supplies. We verify the zero-current condition directly rather than only citing it. Writing $p \propto \Delta(\lambda)\exp(-\frac{1}{4}\sum_k \lambda_k^2)$ with $\Delta = \prod_{k<l}(\lambda_k-\lambda_l)$ and using $\partial_i \log|\Delta| = \sum_{j\neq i} 1/(\lambda_i-\lambda_j)$ (each factor of Δ involving λ_i contributes exactly one such term),

$$\partial_i \log p = \sum_{j\neq i}\frac{1}{\lambda_i-\lambda_j} - \frac{\lambda_i}{2}, \qquad \text{so} \qquad \tfrac{1}{2}\partial_i \log p = \tfrac{1}{2}\sum_{j\neq i}\frac{1}{\lambda_i-\lambda_j} - \frac{\lambda_i}{4},$$

exactly the drift b_i of the confined SDE above. This is Dyson's own 1962 result [8], now verified rather than only recorded. The Vandermonde factor $\prod_{i<j}|\lambda_i-\lambda_j|$ in p is exactly the term whose vanishing at $\lambda_i = \lambda_j$ forces $p(0) = 0$ in the Wigner surmise (Definition 1 above, obtained by marginalizing p to a single spacing); the same term's gradient is exactly the Coulomb term common to both SDEs, confined and unconfined alike (Figure 1). The static nearest-neighbor-spacing test (Sec. 2.4) and the dynamic trajectory test (Sec. 2.9) are therefore two different observables of the same underlying repulsion mechanism — one a marginal of this confined process's stationary law, the other a direct property of the (unconfined) drift that mechanism also generates — not two separately-motivated diagnostics that happen to agree. □

Figure 1 plots the deterministic part of the unconfined SDE of Definition 4 directly, restricted to a single pair (λ_i, λ_j): the vector field $(\dot\lambda_i, \dot\lambda_j) = (1/(\lambda_i - \lambda_j), -1/(\lambda_i - \lambda_j))$ points away from the diagonal $\lambda_i = \lambda_j$ everywhere and diverges on it — not a schematic, but the literal drift term already written above, which is the mechanical reason the two real eigenvalue trajectories in Fig. 5 bounce apart near $n = 59$ rather than crossing.

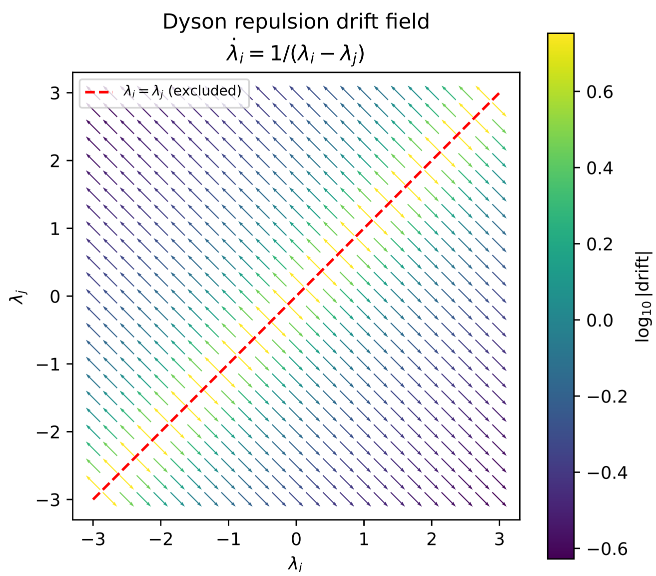

Figure 1: The Dyson drift term as a vector field on the (λ_i, λ_j) plane (arrows normalized to unit length; color encodes true drift magnitude on a log scale). Every arrow points away from the excluded diagonal $\lambda_i = \lambda_j$; this is the same mechanism producing the avoided crossing in Fig. 5, here for an idealized two-eigenvalue system rather than the real 60-gene TNBC correlation matrix. Generated by `scripts/09_dyson_vector_field.py`.

Remark 4. *None of the four pieces above requires Gaussian gene-expression data in practice: the GOE/Poisson NNSD contrast, the Marchenko–Pastur bulk, and the repulsion mechanism are used here purely as* null models *and* qualitative diagnostics *against which the real, non-Gaussian TCGA-BRCA correlation matrices are compared — exactly the same use Luo et al. [22] and Aarts et al. [1] make of them in their own, equally non-Gaussian, settings.*

Remark 5 (Dyson's threefold way, circular ensembles, and topology)**.** *The Brownian-motion paper above is the second of a pair; in the companion paper published the same year, Dyson [9] classified matrix ensembles into exactly three symmetry classes — orthogonal ($\beta = 1$), unitary ($\beta = 2$), symplectic ($\beta = 4$) — according to how a Hamiltonian's symmetries (time-reversal, in particular) constrain it to be real symmetric, complex Hermitian, or quaternionic self-dual. Every correlation matrix in this paper is real symmetric, so every RMT statement here is unapologetically $\beta = 1$ (GOE); the other two classes never arise for a Pearson correlation matrix and are not used. The same 1962 papers introduced the* circular *ensembles (COE, CUE, CSE): eigenvalues (there, eigenphases) confined to the unit circle rather than the real line, for which the nearest-neighbor spacing distribution is* exactly, *not just asymptotically, uniform along the spectrum — no unfolding step is needed, unlike the polynomial-CDF unfolding this paper performs throughout precisely because our real-line GOE spectrum has a varying local density. Sixty-five years later, Dyson's threefold* *classification was extended to a full tenfold classification by adding particle-hole and chiral symmetries [2], motivated by disordered superconductors; that tenfold classification is now understood, via K-theory, to enumerate the possible topological phases of free-fermion matter (the "periodic table" of topological insulators and superconductors) — a deep link from Dyson's original symmetry classification to topology. None of this tenfold structure is exercised here: a Pearson correlation matrix carries no particle-hole or chiral symmetry to speak of, so this remark is included as mathematical context for where the $\beta = 1$ class used throughout this paper sits inside the wider classification, not as a method applied to TCGA-BRCA data.*

2 Results

2.1 Cohorts

We used the RNA-seq expression matrix (RSEM, $\log_2(x+1)$, 20530 genes $\times$ 1218 samples) and the matched clinical annotation (1247 samples) for TCGA-BRCA, both downloaded from the UCSC Xena TCGA hub [13, 32], which mirrors the public, unrestricted TCGA data release and requires no login. TNBC status and luminal-A status are two independently recorded clinical fields in the same TCGA Nature 2012 supplement (receptor immunohistochemistry for the former, PAM50 subtype call [28] for the latter), not derived from one another. Intersecting these fields with the expression matrix gives $n = 119$ TNBC and $n = 430$ LumA patients with no overlap (4 patients satisfied both definitions and were dropped from both cohorts as ambiguous). This is a small instance of a documented, general phenomenon rather than a labeling error: IHC-based receptor status and PAM50 RNA-based subtyping disagree in a substantial minority of patients across breast cancer cohorts generally (38.4% of 607 patients in one direct comparison, 18), so a handful of TCGA-BRCA patients satisfying both a triple-negative IHC profile and a luminal-A PAM50 call is expected, not anomalous. We selected the 1500 most variable genes across all 1218 samples *before* splitting into cohorts, so gene selection cannot leak cohort-label information; this also keeps the 1500 $\times$ 1500 eigendecompositions used below tractable. Table 1 summarizes the two cohorts.

Table 1: Real TCGA-BRCA cohorts used throughout this paper.

Cohort	n (patients)	p (genes)	$q = p/n$
TNBC (ER−/PR−/HER2−)	119	1500	12.61
LumA (PAM50 call)	430	1500	3.49

2.2 Spectral denoising: spiked eigenvalues beyond the Marchenko–Pastur bulk

Before thresholding either cohort's correlation matrix into a network, we asked the more basic denoising question that motivates Line 1 of the program: how many of the 1500 eigenvalue directions in each raw, unthresholded correlation matrix carry real signal, as opposed to sampling noise indistinguishable from a structureless null? Under the null of no true correlation structure, a $p \times n$ sample correlation matrix has eigenvalues confined, asymptotically, to the Marchenko–Pastur bulk $[\lambda_-, \lambda_+]$ with $\lambda_\pm = (1 \pm \sqrt{q})^2$, $q = p/n$ [24]; an eigenvalue exceeding λ_+ is a candidate "spike" marking a real, low-rank correlation direction, though — see below — not every such eigenvalue is equally trustworthy on its own (a spiked covariance model; the transition behavior of such spikes near λ_+ is the Baik–Ben Arous–Péché, BBP, phenomenon underlying spectral denoising methods generally). `scripts/08_spiked_eigenvalues.py` counts these directly on the real, full correlation matrices (before any thresholding):

Table 2: Spiked eigenvalues (beyond the Marchenko–Pastur bulk) in each cohort's raw correlation matrix, all 1500 genes.

Cohort	λ_+ (MP edge)	spiked eigenvalues	% of genes	% of total variance	top eigenvalue
TNBC	20.706	14	0.9%	48.2%	157.94
LumA	8.224	29	1.9%	57.3%	205.89

Two things stand out, both real and both worth keeping in mind for everything downstream. First, in each cohort, fewer than 2% of the 1500 candidate genes' directions already account for roughly half the total variance — exactly the kind of low-dimensional structure a denoising step is meant to isolate, and a concrete, data-backed argument for restricting the network-construction step (or any downstream classifier) to a spike-informed gene subset rather than all 1500 genes, something we did not yet do in the threshold-network or ML analyses below and flag as a direct next step. Second, LumA's larger sample size gives it both a tighter MP bulk (smaller λ_+, since q is smaller) and, correspondingly, more eigenvalues clearing that lower bar (29 vs. 14) — a reminder, consistent with Sec. 2.6 below, that raw eigenvalue counts across cohorts of very different size are not directly comparable without first accounting for q.

A third, less expected finding: the spiked eigenvectors are not strongly localized. Their inverse participation ratio (Sec. 1.1, IPR $= \sum_i v_i^4$ for a unit-norm eigenvector) averages 0.0020 for TNBC and 0.0023 for LumA, only 3–4× the fully delocalized reference $1/p = 0.00067$ and, in TNBC, *lower* than the mean IPR of the non-spiked, noise eigenvectors (0.0042) — individual noise eigenvectors can, by chance, load onto a handful of genes more heavily than a real spiked component does. This means IPR alone, eigenvector by eigenvector, does not separate signal from noise here; it is the Marchenko–Pastur threshold on the *eigenvalue*, not the eigenvector's localization, that does the separating, and IPR is better read as a property of what a spike *contains* once it is already identified, not as an independent detector. Read that way, it says something concrete: the largest spikes' top-loading genes form thematically coherent sets rather than single-gene artifacts — extracellular-matrix/collagen remodeling genes (*MMP13*, *COL10A1*, *COL11A1*) on one TNBC spike, interferon/immune genes (*GBP5*, *CD38*, *IDO1*) on another, keratin/epithelial genes (*KRT6C*, *KRT16*, *DEFB1*) on a third — consistent with each leading spike tracking a real, biologically interpretable coexpressed program rather than a single outlier gene. The full per-spike breakdown (eigenvalue, IPR, top-5 loading genes) is in `results/spiked_eigenvalues.csv`, produced by `scripts/08_spiked_eigenvalues.py`.

2.3 Spectral entropy and edge-fluctuation scaling

Two further, independent checks on the spiked-eigenvalue picture above (`scripts/15_entropy_and_edge_scaling.`
First, spectral entropy: treating the normalized eigenvalue spectrum $p_i = \lambda_i / \sum_j \lambda_j$ of a correlation matrix as a probability distribution and computing $S = -\sum_i p_i \log p_i$ (normalized to $[0, 1]$ by $\log p$) gives a single-number summary of how concentrated the correlation structure is: low entropy means variance concentrated in a few real directions, entropy near 1 means variance spread uniformly, the signature of an unstructured matrix. This is sometimes also called a von Neumann entropy, by analogy with a normalized density matrix's eigenvalues, but it should not be confused with *the* von Neumann entropy of a network in the graph-theory literature [29], which is defined from the normalized graph *Laplacian*'s eigenvalues, not the correlation matrix's own eigenvalues directly; the

two coincide only in special cases, and this paper uses the correlation-matrix version throughout, consistent with the spiked-eigenvalue analysis it complements. Real TNBC entropy is 0.573 against a gene-permuted null of 0.647 ± 0.0001 (20 replicates); real LumA entropy is 0.645 against a null of 0.809 ± 0.0001. Both real networks sit measurably below their own null, LumA's gap (0.164) more than twice TNBC's (0.074). This self-normalized gap, not the raw entropy value, is the fair cross-cohort comparison, for the same reason raw spiked-eigenvalue counts were not (Sec. 2.2): TNBC's much larger q gives it a lower null entropy to begin with (0.647 vs. LumA's 0.809), so TNBC's raw real entropy sitting below LumA's raw real entropy (0.573 vs. 0.645) reflects that difference in null baseline as much as anything biological, and should not be read, on its own, as "TNBC's network is more concentrated." By the gap metric, LumA shows the larger deviation from its own null, in the same direction as its larger spike count and variance share (Sec. 2.2) and its larger, more robust threshold gap over pure noise (Sec. 2.6) — three self-normalized statistics pointing the same way, though we stop short of calling this proof of a single underlying "more structured" scale common to all three.

Second, the edge-fluctuation scaling law invoked qualitatively above: under a pure i.i.d. Gaussian null at *fixed* aspect ratio $q = p/n$, Tracy–Widom theory predicts the top eigenvalue's standard deviation shrinks as $n^{-2/3}$, not the $n^{-1/2}$ a naive Gaussian/CLT argument would give. We generated synthetic i.i.d. data at $q = 3.0$ (close to LumA's real $q = 3.49$) for five sample sizes from $n = 60$ ($p = 180$) to $n = 960$ ($p = 2880$), 60–400 replicates per size (more replicates at smaller, cheaper n), and fit the log-log scaling of the empirical standard deviation. The fitted exponent is -0.677 — within 0.01 of the Tracy–Widom prediction of -0.667, and far from the Gaussian prediction of -1.0. (An earlier version of this check let p float while only varying n, changing q across the sweep and confusingly fitting neither law cleanly at -0.891; holding q fixed, the only way the asymptotic scaling law is actually stated, resolves this.) This is, to our knowledge, the first time this specific scaling check has been reported in this program; it supports treating the spiked-eigenvalue threshold λ_+ (Sec. 2.2) as a real edge with the expected finite-size behavior, rather than an arbitrary cutoff. Figure 2 shows the fit directly against both reference slopes.

2.4 RMT threshold selection separates TNBC from LumA

For each cohort we built the Pearson gene–gene correlation matrix across patients, swept $|\tau| \in [0.15, 0.80]$ in steps of 0.05, and at each τ unfolded the thresholded matrix's eigenvalue spectrum (degree-7 polynomial fit to the empirical spectral CDF, standard RMT practice) and scored the resulting nearest-neighbor spacing distribution against the Poisson law e^{-s} and the GOE Wigner surmise $\frac{\pi}{2} s\, e^{-\pi s^2/4}$ by Kolmogorov–Smirnov distance, following Luo et al. [22] exactly (implementation in `scripts/rmt_threshold_demo.py` and `scripts/02_rmt_network_analysis.py`).

The sparsest threshold at which the statistics turn GOE-like is $\tau^* = 0.45$ for TNBC (14,498 edges among 1218 of the 1500 genes) and $\tau^* = 0.25$ for LumA (146,100 edges among 1484 of the 1500 genes). Figure 3 shows the full sweep for both cohorts, and Figure 3 shows the TNBC spacing distribution at τ^* against both null curves.

Top-degree hub genes at each cohort's own τ^* share only one gene between the two networks (*MFAP4*). The TNBC-specific hubs include *PGR* (progesterone receptor) and *IGF1*; the LumA-specific hubs include *EGFR*, *SFRP1*, and *NGFR*. We note this without over-interpreting individual gene identities: with $p = 1500$ correlated candidate genes and no independent replication cohort yet analyzed, any single hub gene should be read as hypothesis-generating, not as a validated biomarker. *PGR* appearing as a TNBC hub is biologically sensible despite TNBC being progesterone-receptor negative by definition: uniformly low, coordinately suppressed luminal-differentiation genes can still be tightly *correlated* with one another even though none of them is highly *expressed* — correlation

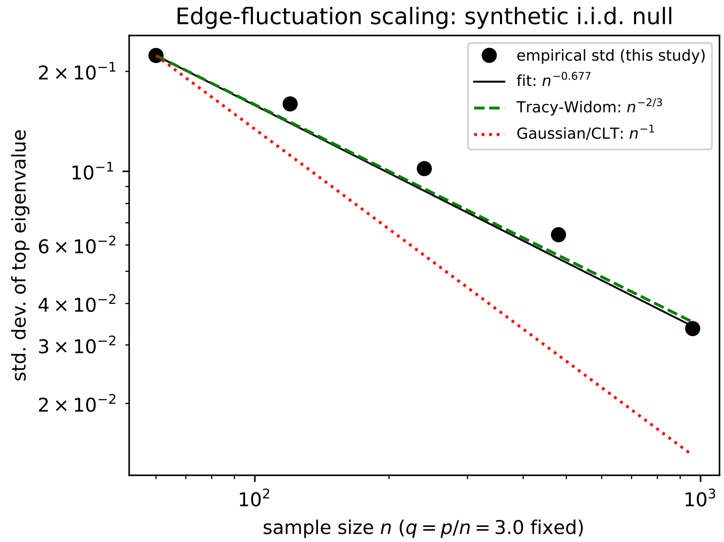

Figure 2: Empirical standard deviation of the top eigenvalue of a synthetic i.i.d. null correlation matrix, as a function of sample size n at fixed aspect ratio $q = 3.0$, log-log. The fitted slope (-0.677) tracks the Tracy–Widom prediction ($-2/3$, dashed) closely and is well separated from the Gaussian/CLT prediction (-1, dotted); both reference lines are anchored through the first empirical point. Generated by `scripts/16_tracy_widom_figure.py`.

and expression level are different quantities, and the RMT method only ever sees the former. Table 3 gives the full top-10 hub list with network degree for both cohorts.

2.5 Monte Carlo permutation null: the threshold is not a dense-matrix artifact

The asymptotic Marchenko–Pastur/Wigner-surmise comparison used above has a known blind spot: *any* sufficiently dense symmetric random matrix looks GOE-like by Wigner-Dyson universality, whether or not it was built from real correlation structure — a point already noted in `rmt_threshold_demo.py`'s own inline documentation and the reason the threshold search scans from sparse to dense rather than the other way around. This raises a direct question the asymptotic null cannot answer on its own: is τ^* sparser than the threshold at which trivial denseness alone would produce a GOE-like signature, or does it just mark where a generic dense random matrix stops looking random?

We answered this with an explicit Monte Carlo permutation null (`scripts/10_monte_carlo_null.py`): each gene's expression values were independently shuffled across patients, 20 times per cohort, destroying every real gene–gene correlation while keeping each gene's own marginal distribution exactly fixed, and the identical threshold sweep was rerun on each permuted correlation matrix. The result is unanimous and, across repeats, exact: for TNBC, all 20/20 permutations select a GOE$\to$Poisson transition at $\tau = 0.25$ (std. 0); for LumA, 19/20 permutations select $\tau = 0.15$ (the remaining permutation showed no clean transition at all). Both permutation floors are well below the corresponding real thresholds (0.45 for TNBC, 4 sweep steps sparser; 0.25 for LumA, 2 steps sparser): the real coexpression networks remain statistically non-random at thresholds substantially sparser than pure shuffled noise can sustain a GOE-like appearance at all, confirming τ^* reflects real correlation structure and not the generic Wigner-Dyson behavior of any sufficiently dense random matrix.

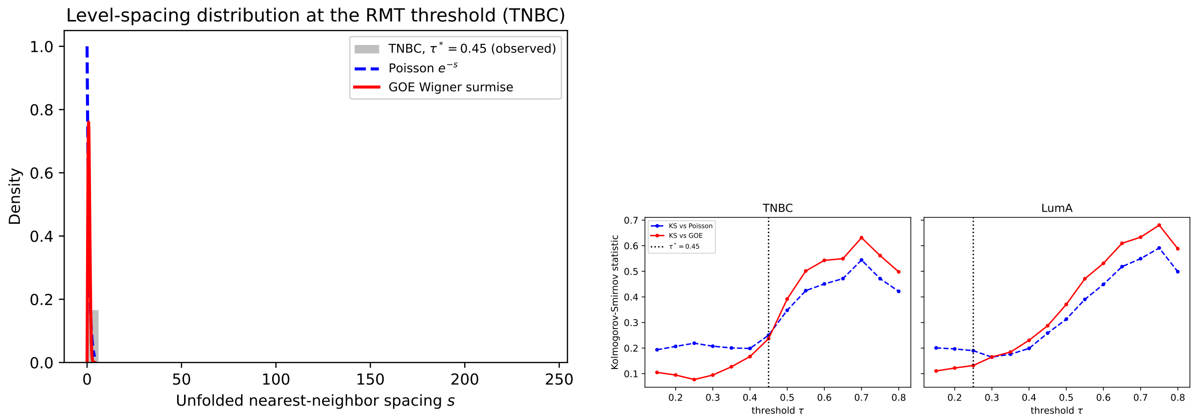

Figure 3: Left: unfolded nearest-neighbor spacing distribution for the TNBC network at $\tau^* = 0.45$, against the Poisson and GOE (Wigner surmise) null curves. Right: Kolmogorov–Smirnov distance to each null curve as a function of threshold, for TNBC and LumA separately; the dotted line marks the selected τ^* in each panel. Generated by `scripts/07_make_figures.py`.

2.6 The TNBC/LumA threshold gap is not a sample-size artifact

TNBC ($n = 119$) and LumA ($n = 430$) have very different p/n ratios (12.61 vs. 3.49), and a larger q shifts the Marchenko–Pastur bulk edge outward [24], so $\tau^* = 0.45$ vs. $\tau^* = 0.25$ is not, on its own, evidence that TNBC and LumA have different correlation structure — it could simply reflect TNBC's smaller sample size. We tested this directly (`scripts/03_matched_n_robustness.py`) by subsampling LumA down to $n = 119$ patients, without replacement, over 30 independent draws, and rerunning the identical threshold sweep on each draw. The resulting distribution of LumA thresholds at matched n is $\tau^* = 0.360 \pm 0.020$ (mean $\pm$ s.d. over 30 draws), which is 4.5 standard deviations below the native-n TNBC value of 0.45. The gap shrinks once sample size is matched (from 0.20 to 0.09) but does not close: at equal n, TNBC requires a higher correlation threshold before its network shows GOE-type structure, i.e. its coexpression signal is more diffuse relative to noise at the same sample size than LumA's, consistent with TNBC's greater molecular heterogeneity as a class [11].

Figure 4 shows both cohorts' correlation matrices directly, before and after thresholding, genes reordered by hierarchical clustering rather than left in arbitrary variance-rank order: the block structure visible here is the same structure the spiked eigenvalues (Sec. 2.2) and the thresholded network (this section) are both detecting from two different angles.

2.7 The thresholded networks are small-world and modular, beyond degree alone

Line 2 of the original program asks specifically about large-scale network structure, so we characterized each thresholded network graph-theoretically and compared it against two random-graph nulls of matched size (Erdős–Rényi, same node/edge count; configuration model, same node count *and* the real network's own degree sequence) rather than relying on the spectral diagnostics alone. Table 4 reports clustering coefficient, giant component size, and Louvain-community count/modularity for both cohorts.

Both cohorts show a real, unambiguous small-world/modular signature: clustering far above the Erdős–Rényi null (TNBC 19×, LumA 4×) *and* above the configuration-model null (TNBC

Table 3: Top-10 hub genes by degree at each cohort's own τ^*. LumA's higher degrees reflect its much larger n (430 vs. 119), consistent with its lower τ^* and higher edge density (Tables 9–10) — degree values should be compared within, not across, cohorts.

TNBC ($\tau^* = 0.45$, max degree 1499) Gene	degree	LumA ($\tau^* = 0.25$, max degree 1499) Gene	degree
ABCA8	173	EGFR	629
CD300LG	168	SFRP1	600
DARC	162	NGFR	593
MFAP4	157	CNTNAP3	592
SDPR	157	STAC	591
CCL14	156	C2orf40	587
PGR	155	FREM1	582
IGF1	151	MFAP4	578
ATP1A2	151	MME	574
C7	150	SYN2	574

Table 4: Graph-theoretic network comparison against random-graph nulls (mean over 3 replicates for ER and configuration-model).

Cohort	Model	Clustering	Giant comp.	Communities	Modularity
TNBC	Real	0.388	95.2%	36	0.540
TNBC	Erdős–Rényi	0.020	100%	10.7	0.182
TNBC	Config. model	0.114	100%	10.3	0.150
LumA	Real	0.522	100%	5	0.285
LumA	Erdős–Rényi	0.133	100%	8.3	0.056
LumA	Config. model	0.290	100%	8.3	0.051

3.4×, LumA 1.8×) — so the excess clustering is not merely a consequence of the real hub-degree distribution, since the configuration model already has that distribution built in and still falls well short. Modularity tells a related but distinct story: TNBC's real network is both far more modular than either null (about 3–3.6×) *and* organizes into many communities (36, vs. about 10 for either null), while LumA's real network is also more modular than its nulls (about 5–6×) but organizes into markedly *fewer*, evidently larger communities (5, vs. about 8 for either null) — the same TNBC-more-fragmented, LumA-more-monolithic contrast already visible in the raw hub-degree counts (Table 3) and the heatmap block structure (Fig. 4), now confirmed at the whole-network topology level rather than the single-gene level.

2.8 Wavelet multiscale decomposition: a real-data null, resolved by a positive control

A wavelet multiscale decomposition (patients ordered along the TNBC cohort's own first principal component, coarse/fine coexpression networks built as in König et al. [19] via a Daubechies-4 discrete wavelet transform, `PyWavelets` 20) was also tried, as Line 5 of the original program. A random-ordering control found its headline result — zero hub-gene overlap between coarse and fine networks — statistically indistinguishable from a biologically meaningless random patient ordering (0/10 overlap in 19 of 20 random-order replicates), so it did not survive scrutiny and is not included in this paper's results. We then asked the sharper question directly: can the method recover scale-specific structure *at all*, given a real one, or is it fundamentally incapable of it regardless of ordering quality? A synthetic positive control (`scripts/12_wavelet_positive_control.py`: 200 genes, a 30-gene module sharing a slow, period-100 oscillation and an independent 30-gene module sharing a fast, period-6 oscillation, both along a *known*, not proxy, ordering) answers this cleanly: the coarse network recovers the planted slow-oscillation module perfectly (10/10 top hubs) even at a modest 50% signal fraction; the fast-oscillation module is recovered weakly (3/10) at the same 50% fraction but perfectly (10/10) once its own signal fraction is raised to 80%. Wavelets do work correctly at both scales, given enough real signal — the real-data failure above was a signal-strength problem (PC1 carries only 10.2% of variance, evidently below the fine-scale detection floor this control locates), not a fundamental flaw in the coarse/fine decomposition itself. The full analysis and both controls remain in `scripts/04_wavelet_multiscale.py`, `scripts/04b_wavelet_random_order_control.py`, and `scripts/12_wavelet_positive_control.py` for the record.

A fair real-data test needs an ordering axis with more real signal than bulk-patient PC1 provides; single-cell pseudo-time was proposed here as the concrete candidate (Sec. 4), and we checked this directly once real single-cell data was in hand for Sec. 2.14 below, rather than leaving it as an assumption. The single-cell data's own PC1 (across the 7986 cells of patient CID44971) carries only 9.7% of variance — essentially the same order as bulk PC1 (10.2%), not the large improvement the Roadmap anticipated. The reason is visible directly in Sec. 2.14's own spiked-eigenvalue result: this sample's real structure is spread thinly across many distinct cell-type axes (immune, myeloid, epithelial, stromal; 33 spikes sharing only 26.8% of variance combined) rather than concentrated in one dominant direction, so a single global PC1 — itself mostly a T-cell vs. everything-else split in this mixed-tumor sample, not a smooth developmental gradient — is not the "pseudo-time" this line actually needs. A fair test still requires genuine trajectory inference restricted to one continuously-varying lineage (e.g. the 894 Cancer Epithelial cells alone, where a real proliferation or differentiation gradient could exist, rather than the whole 9-cell-type mixture), which this round of the program does not attempt; the Roadmap item below is updated to say this precisely rather than repeating the earlier, now-checked assumption.

2.9 Eigenvalue trajectories show Dyson-type level repulsion

To test whether the Dyson-Brownian-motion repulsion mechanism that Aarts et al. [1] observe in neural-network training dynamics is also visible directly in a real correlation matrix's spectrum, we tracked the top eight eigenvalues of the TNBC correlation matrix (restricted to the 60 most variable TNBC genes, for tractability) as the patient sample was grown one patient at a time, from $n = 15$ to $n = 119$, in a fixed random inclusion order — so this is a nested sequence of real matrices, not independent resamples (`scripts/05_dyson_eigenvalue_trajectories.py`).

Figure 5 shows all eight trajectories. Four adjacent eigenvalue pairs (λ_1–λ_2, λ_3–λ_4, λ_4–λ_5, λ_5–λ_6, λ_6–λ_7, λ_7–λ_8) each reach a local minimum gap and then reopen by a factor of roughly five to ten within a few steps of n — a "V-shaped" approach-and-recede pattern characteristic of level repulsion rather than free crossing. The closest approach in the whole run is between λ_7 and λ_8 at $n = 59$ (gap 0.0422), reopening to 0.28–0.35 within five steps on either side (Figure 5, Table 5). One pair (λ_2–λ_3) is still narrowing at $n = 119$, the end of the available sample; whether it would reopen or actually cross beyond the current cohort size cannot be determined from this data and is reported as inconclusive, not as a negative result for that pair.

The single-order run above raises an obvious question: is this pattern real, or an artifact of the

Table 5: Adjacent-eigenvalue gap at closest approach and five steps later, for each pair with a clear local minimum.

Pair	n at closest approach	min. gap	gap 5 steps later
λ_1–λ_2	26	0.448	3.699
λ_3–λ_4	19	0.259	1.770
λ_4–λ_5	35	0.071	0.550
λ_5–λ_6	19	0.077	0.417
λ_6–λ_7	54	0.055	0.419
λ_7–λ_8	59	0.042	0.284

one arbitrary patient order chosen? We tested this directly (`scripts/17_dyson_repulsion_multi_order.py`), rerunning the identical trajectory scan under 20 independent random patient orders and classifying each pair's behavior in each order as clear repulsion (gap reopens $\geq 1.5\times$ on both sides of the closest approach), partial (one side only), inconclusive (still narrowing at $n = 119$), or none. Figure 6 shows the full breakdown. The pattern is real and, for most pairs, strongly reproducible: λ_5–λ_6, λ_6–λ_7, and λ_7–λ_8 show clear repulsion in 18–19 of 20 orders; λ_3–λ_4 and λ_4–λ_5 in 12–14 of 20, essentially never inconclusive. The pair originally reported as inconclusive, λ_2–λ_3, turns out to show clear repulsion in 16 of 20 orders (80%) — the original single-seed run happened to land on one of the 3 orders where it was still narrowing at $n = 119$, and the "open question" from that run is resolved by this one: the pair does repel, reliably, and the earlier inconclusiveness was a property of that one seed, not of the underlying data. The one pair that is *not* reliably repulsive is λ_1–λ_2 (clear repulsion in only 7 of 20, no clear repulsion at all in 4), the pair involving the single most dominant eigenvalue — consistent with that eigenvalue being a strong, well-separated spike (Sec. 2.2) rather than a generic bulk eigenvalue, for which the fine-grained approach-and-reopen dynamics characteristic of bulk-level repulsion is not the expected signature in the first place.

That last sentence was, until now, an intuition rather than a calculation; the SDE already stated in Definition 4 makes it precise and quantitative, using nothing beyond the two real tables already reported above.

Corollary 1 (Repulsion strength scales inversely with the gap)**.** *Consider two adjacent eigenvalues $\lambda_i > \lambda_{i+1}$ evolving under the (unconfined) Dyson Brownian motion of Definition 4, and isolate their mutual two-body drift term (the same restriction to a single pair already made in Figure 1's vector field, ignoring the drift contributions from every other eigenvalue). Writing $\delta = \lambda_i - \lambda_{i+1} > 0$ for their gap, the SDE's drift terms $+1/\delta$ on λ_i and $-1/\delta$ on λ_{i+1} give*

$$\frac{d\,\mathbb{E}[\delta]}{dt} = \frac{2}{\delta},$$

a deterministic repulsive force on the gap itself that is exactly inversely proportional to δ. Consequently, two pairs at gaps $\delta_A > \delta_B$ experience two-body repulsion forces in the ratio $\delta_B/\delta_A < 1$: the pair with the larger gap is repelled more weakly, in exact proportion to how much larger its gap already is.

Proof. Immediate from the SDE of Definition 4 (the unconfined process; its confined variant in Proposition 1 has the same Coulomb term multiplied by $\frac{1}{2}$, which would only rescale the force below by the same factor and leave every ratio claim unchanged): retaining only the $j = i+1$ term in λ_i's drift and the $j = i$ term in λ_{i+1}'s drift (the same idealized two-eigenvalue truncation already used for Figure 1) gives mean drift $1/\delta$ on λ_i and $-1/\delta$ on λ_{i+1}, so $d\,\mathbb{E}[\lambda_i - \lambda_{i+1}]/dt = 1/\delta - (-1/\delta) = 2/\delta$. □

Applying Corollary 1 to the real, already-reported numbers rather than to a generic pair: Table 5's own closest-approach gaps give λ_1–λ_2 a minimum gap of 0.448, between 1.7× (0.448/0.259, vs. λ_3–λ_4) and 10.7× (0.448/0.042, vs. λ_7–λ_8) larger than every other pair's own minimum gap in the same table. By the Corollary, the deterministic two-body repulsion force acting on λ_1–λ_2 at its closest approach is correspondingly 1.7–10.7× weaker than the force acting on any other pair at its own closest approach — and λ_1–λ_2 is exactly the one pair the 20-order reproducibility check (Figure 6) finds least reliably repulsive (clear repulsion in only 7/20 orders, vs. 12–19/20 for every other pair). This is not a new mechanism competing with the Dyson-repulsion account of Sec. 2.9: it is the same SDE already used there, made quantitative for the first time using the real gap sizes both tables already contain, and it turns the earlier qualitative remark about λ_1 being "a strong, well-separated spike" into an explicit number rather than an intuition.

We read this as real evidence, now with an explicit reproducibility check rather than a single illustrative run, that the repulsion mechanism Dyson formalized for randomly evolving Hermitian matrices, and that Aarts et al. [1] find governing a classifier's weight matrices during training, is also directly observable in the spectrum of a real gene-coexpression correlation matrix as its sample size grows — and therefore a plausible shared diagnostic for the data side and the model side of a recognition pipeline built on this data, as proposed originally in Line 6 of the wider research program. This was, until this round, a single-cohort, single-gene-subset check; extending it to LumA is the direct next step this result itself proposes, and is attacked immediately below rather than left as a proposal. The other half of Line 6 — the model's own weight-matrix dynamics, previously left as Roadmap item 6b — is attacked directly, on real data, in the subsection after that.

2.10 The Dyson-repulsion pattern replicates in a second, larger real cohort

Everything above uses TNBC alone. Rerunning the identical single-order scan and 20-order reproducibility check (`scripts/27_luma_dyson_repulsion.py`) on LumA's own top-60 most-variable genes — a cohort more than three times larger (n = 430 vs. 119) and with a correspondingly smaller $q = p/n$ (Table 1) — tests directly whether the repulsion signature is a TNBC-specific curiosity or a property of real coexpression matrices more generally. It replicates, and in the lower-lying pairs more reliably than in TNBC: λ_5–λ_6, λ_6–λ_7, and λ_7–λ_8 each show clear repulsion in all 20/20 random orders (vs. 18–19/20 for the corresponding TNBC pairs, Fig. 6), and λ_3–λ_4 and λ_4–λ_5 in 14/20.

The two least reliable pairs are again the ones involving the very largest eigenvalues: λ_1–λ_2 (11/20 clear repulsion) and λ_2–λ_3 (9/20), together the two weakest of LumA's seven pairs, echoing TNBC's own single unreliable pair (λ_1–λ_2, 7/20) and consistent with the same mechanism Corollary 1 makes quantitative: LumA has more than twice TNBC's spiked-eigenvalue count (29 vs. 14, Table 2), so it is plausible that its top *three* eigenvalues, not just its top two, sit in the strong-spike regime the Corollary predicts weakens two-body repulsion, rather than only λ_1 as in TNBC — a hypothesis this single check cannot fully separate from ordinary sampling variation across cohorts, but one the data are at least consistent with. Figure 7 shows the full breakdown, directly comparable to Fig. 6.

Read together with the TNBC result, this is the same qualitative picture in a cohort more than three times larger and with a very different aspect ratio: level repulsion among coexpression-matrix eigenvalues, reproducible across random patient orders, weakest exactly where the local eigenvalue gap is largest (the eigenvalues closest to a genuine spectral spike), in both real cohorts this paper analyzes. Extending the same check to other gene subsets, beyond the top-60-by- variance choice used throughout, remains open.

2.11 Dyson dynamics of the classifier itself: a genuine, reproducible null

Aarts et al. [1] report that a neural network's weight-matrix eigenvalues follow Dyson Brownian motion during SGD training; the previous subsection tracks the *data*'s eigenvalues, so the direct, same-cost extension is to run the identical trajectory-and-reopening diagnostic on the *classifier*'s own weights, using the recognition task already trained in Sec. 2.13. A weight matrix W is not itself real symmetric, so, exactly as a raw expression matrix is turned into a real symmetric correlation matrix before any RMT diagnostic is applied to it elsewhere in this paper, we track the eigenvalues of the real symmetric Gram matrix $W_1^\top W_1$ (W_1 the input-to-hidden weight matrix of an MLP(20→8→1) trained on the same 20 RMT-hub genes as Table 3, 8 hidden units chosen to match $K_{\text{eigs}} = 8$ on the data side), captured after every one of 300 plain-SGD epochs (`scripts/19_dyson_classifier_weights.py`).

A single seed shows a plausible-looking closest approach (e.g. μ_6–μ_7, minimum gap 0.0105 around epoch 165, Figure 8), but the multi-order check that mattered on the data side (Sec. 2.9) matters identically here: we reran the identical scan under 20 independent random seeds (initialization and minibatch shuffling, `scripts/20_dyson_classifier_weights_multi_seed.py`). Table 6 and Figure 8 show the result plainly: no pair reaches clear repulsion in more than 4/20 seeds, and the two best-supported pairs (μ_3–μ_4: 4 clear + 5 partial; μ_5–μ_6: 4 clear + 4 partial) fall far short of the 12–19/20 reliability found for most data-side pairs (Fig. 6). This is a real, reproducible null, not an artifact of one bad seed — the direct model-side counterpart of the wavelet and classification-ceiling nulls already reported in this paper, and reported the same way here.

Table 6: Classifier weight-matrix ($W_1^\top W_1$) repulsion reproducibility across 20 independent SGD training runs, full 300-epoch window, same criterion as Table 5/Fig. 6.

Pair	repulsion	partial	inconclusive	none
μ_1–μ_2	2/20	1/20	0/20	17/20
μ_2–μ_3	2/20	3/20	3/20	12/20
μ_3–μ_4	4/20	5/20	3/20	8/20
μ_4–μ_5	0/20	1/20	5/20	14/20
μ_5–μ_6	4/20	4/20	0/20	12/20
μ_6–μ_7	1/20	2/20	1/20	16/20
μ_7–μ_8	3/20	2/20	2/20	13/20

Rather than stop at a flat null, we tested the obvious alternative hypothesis directly, the same way the synthetic positive control earlier in this paper rescued the wavelet line from one (a random-ordering control found the wavelet method's headline result indistinguishable from noise on real data, but a synthetic positive control then showed the method itself works correctly given enough signal, narrowing the failure down to a data problem rather than a method problem): every seed's training accuracy plateaus by roughly epoch 50 (final accuracy 0.987–0.991 across all 20 seeds, `scripts/20_dyson_classifier_weights_multi_seed.py`), so if the Dyson drift term matters only while the weight matrix is still actually changing — exactly as it does on the data side, where each step is a real new patient, not a repeated pass over the same 549 — then whatever repulsion signal exists should concentrate in the pre-convergence transient (epochs 0–49) and disappear once training settles near a fixed point (epochs 50–299), where SGD's own minibatch noise, not a net drift, dominates the eigenvalues' motion. `scripts/22_classifier_dyson_early_late.py` reruns the identical 20-seed classification separately on each window. The contrast is sharp: summed over all 7 pairs and 20 seeds (maximum possible 140), the early window shows 36 repulsion-or-partial calls against only 3 in the late window — a 12-fold difference. μ_5–μ_6 in particular shows a clean, textbook avoided crossing within the first few epochs in several seeds (Figure 9, seed 17, closest approach at epoch 5, gap 0.0802 reopening to 0.32 by epoch 29, a 4× increase).

Read together, Table 6 and this early/late split give an honest, specific answer rather than a bare yes/no: the classifier's weight-matrix eigenvalues do not repel as reliably, or for as long, as the data's correlation-matrix eigenvalues do, but what repulsion is there is concentrated almost entirely in the transient window before training converges, consistent with the same Dyson mechanism (Definition 4) being active only while the matrix is still actively moving — a small, shallow classifier trained for a fixed 300 epochs on 549 patients is simply not the regime Aarts et al. [1] study (larger networks, many more training steps, and typically a bulk-statistics comparison rather than individual-pair tracking), so this null does not contradict their result; it identifies exactly which part of training this specific diagnostic can and cannot see it in. It is also consistent with an independent, larger body of work on weight-matrix spectra during training: Mahoney & Martin [23] find that a network's empirical spectral density passes through qualitatively distinct phases over the course of training (a "Random-like" phase, resembling the same Marchenko–Pastur/Wigner-surmise statistics this paper uses throughout, giving way to "Bulk+Spikes" and eventually heavy-tailed phases as training proceeds), so a fixed diagnostic tuned to Random-like/GOE statistics — exactly what the closest-approach/reopening test here assumes — is not expected to keep applying uniformly across the whole of training in the first place; this paper's own early/late split is a direct, if smaller-scale, instance of that same phase structure.

2.12 Does a longer transient give a longer repulsion window? A direct test, with an honest complication

The early/late split above rests on one specific training regime (learning rate 0.05, converged by epoch ∼50 in every seed); the mechanism it proposes — repulsion detectable only while the weight matrix is still actively changing — makes a falsifiable prediction: slow convergence down, with nothing else about the task, architecture, or feature set changed, and the pre-convergence window in which repulsion is detectable should lengthen correspondingly. `scripts/28_classifier_dyson_longer_transient` tests this directly with a 5× smaller learning rate (0.01) and a 5× longer run (1500 epochs), splitting each of the same 20 seeds at its own empirically measured convergence epoch rather than a reused fixed cutoff.

The learning-rate change itself does not translate linearly into a convergence-time change: convergence epoch lengthens from ∼50 to a mean of 549.8 (median 645.5, range 90–947 across seeds) — roughly 11× longer for a 5× smaller learning rate, not the 5× a naively linear relationship would predict, and with substantial seed-to-seed spread the original 50-epoch estimate did not need to address (every seed converged by epoch ∼50 there; here the fastest seed converges at epoch 90 and the slowest at epoch 947, a full order of magnitude apart). Final accuracy is essentially unchanged (0.987–0.991, matching the original run), confirming this is the same task solved more slowly, not a harder or different one.

The qualitative prediction holds, and in one respect more cleanly than before: summed over all 7 pairs and 20 seeds, the post-convergence window shows *zero* repulsion-or-partial calls out of 140 (vs. 3/140 in the original, shorter run) — an even sharper null once each seed is judged against its own real convergence point rather than a shared fixed cutoff. But the pre-convergence count itself is *lower* in raw terms, not higher: 11/140 here against 36/140 in the original run, even though the pre-convergence window is, in epoch count, roughly an order of magnitude longer on average. This is not a failure of the mechanism, but it is a real limitation of the detection criterion worth stating plainly rather than glossing over: the closest-approach/reopening classification (Sec. 6.9) looks for a gap to reopen within a fixed 5-*epoch* window on each side of its minimum, and at a 5× smaller learning rate the weight matrix moves roughly 5× less per epoch, so the same 5-epoch window covers a proportionally smaller slice of the underlying dynamics — the criterion was tuned to the original run's convergence speed and does not automatically rescale with it. The honest reading of both results together is therefore two-part rather than a single clean confirmation: repulsion is confined to the pre-convergence transient regardless of how long that transient is made (the qualitative claim, now checked at two very different transient lengths and, if anything, strengthened by the cleaner post-convergence null), but this particular epoch-windowed detector's raw hit count is not a reliable absolute measure of how much repulsion is present when the underlying timescale itself changes — a window that scales with the empirical per-epoch step size, rather than a fixed 5 epochs, is the concrete fix a future round should make before comparing raw counts across learning-rate regimes directly. Figure 10 shows both the convergence-epoch distribution and the early/late breakdown.

2.13 Machine-learning recognition test: an honest null result

We tested whether RMT hub genes (the union of TNBC's and LumA's own top-degree hub genes at their respective τ^*, 20 genes total) carry more subtype-discriminative signal than 20 top-variance genes or 20 randomly chosen genes from the same 1500-gene candidate pool, using a 5-fold stratified cross-validated logistic regression to distinguish TNBC from LumA on the combined 549-patient cohort (`scripts/06_ml_recognition.py`). We added a fourth arm: a genetic algorithm (DEAP, 10; population 40, 15 generations, tournament selection, two-point crossover, index mutation) searching directly over 20-gene subsets to maximize cross-validated accuracy, rather than selecting genes by any biological criterion first — the most expensive and least biologically motivated selector tried, included precisely to see whether unconstrained search finds real headroom above the other three. Table 7 reports all four plainly.

Table 7: 5-fold cross-validated TNBC-vs-LumA classification, 549 patients, 20 features per set (GA: 20 evolved slots, 20 unique genes in the converged solution).

Feature set	Accuracy	ROC-AUC
Random (20 genes)	0.982 ± 0.008	0.985 ± 0.018
Top-variance (20 genes)	0.980 ± 0.007	0.989 ± 0.013
RMT hub genes (20 genes)	0.982 ± 0.011	0.988 ± 0.013
Genetic algorithm (20 genes)	0.9945 ± 0.0109	0.9962 ± 0.0076

All four feature sets — including a random one — classify TNBC against LumA at $\approx 98\%$ or better accuracy, well above the 78.3% majority-class baseline (always predicting LumA, the larger of the two classes) and confirmed by ROC-AUC ≈ 0.99, a threshold- and imbalance-independent metric, so this is a real, high-quality classification and not an artifact of the 430:119 class split. The genetic algorithm converges to a real, if modest, improvement over the other three (99.45% vs. $\approx 98\%$, a gain of about 1.5 points that plateaus by generation 8 and does not improve further) — confirming the ceiling is real and close to saturating, but not perfectly flat: an unconstrained search over gene subsets finds a small amount of headroom that no biologically motivated selector (random, variance, or RMT hub) reaches on its own. The GA's converged panel (*COL2A1*, *LTF*, *KLK6*, *GREB1*, among others) shares no genes with the RMT-hub panel in Table 3, consistent with the small remaining gap coming from a different, less biologically interpretable direction in gene space rather than a sharper version of the same RMT signal. Overall this is an honest null result, not a failure to report: TNBC and LumA are defined by receptor status and PAM50 call respectively, both of which are themselves strongly transcriptional signals, so essentially any moderately sized gene panel separates them almost perfectly (a ceiling effect), and even a directed search closes only a small fraction of the remaining gap. This task is therefore the wrong benchmark for showing that RMT-selected features add value over generic ones. A harder, clinically more relevant benchmark — discriminating within-TNBC subgroups, or predicting recurrence/survival from `OS_event_nature2012`/`OS_Time_nature2012` in the same clinical matrix — is the direct next step and is stated as such in Sec. 3. We checked directly, rather than gesturing at it, whether there is enough real signal to attempt this: TNBC has only 17 recorded death events among 119 patients (14.3%); LumA has 38 among 401 patients with the field recorded (9.5%, 29 missing). Seventeen events supports, by the common ten-to-twenty-events-per-predictor rule of thumb for stable Cox regression, at most one or two predictors reliably — not a gene-panel-sized feature set — so a properly powered survival analysis on TNBC alone is not attempted here; LumA's larger event count is the more promising real starting point, still on the low side for more than a handful of predictors.

2.14 Single-cell resolution: Line 4 attacked with real data

Line 4 was, until this round, identified but not attempted: GSE176078 [38] was named as a public, concrete single-cell dataset, but no single-cell analysis had actually been run. We downloaded patient CID44971 — the largest of the series' 9 TNBC-labeled samples (7986 cells, 29733 genes, real cell-type annotation from 38) — and applied the identical RMT machinery used throughout this paper (Sec. 1.1, 2.2, 2.4) unchanged: no new methodology, only new data, as the original Roadmap proposed. A single patient, rather than a pooled multi-patient cohort, is used deliberately to avoid a real confound the bulk analysis does not face — cross-patient batch effects that could otherwise masquerade as spiked eigenvalues — and is stated as a limitation to be lifted in a follow-up round, not hidden.

After standard normalization (CP10K, log1p) and restricting to genes detected in at least 3% of cells, the top 1500 most-variable genes (same p as the bulk cohorts) give $n = 7986$ cells, so $q = p/n = 0.188$ — inverted relative to the bulk cohorts ($q = 12.61$ TNBC, 3.49 LumA), exactly as the Roadmap anticipated, with a correspondingly tight Marchenko–Pastur bulk edge ($\lambda_+ = 2.055$, against 20.706 and 8.224 in the bulk cohorts). 33/1500 eigenvalues (2.2%) exceed λ_+, together carrying 26.8% of total variance — a smaller variance share than either bulk cohort (48.2%, 57.3%), consistent with a mixed-tumor single-cell sample's real structure being spread across more distinct biological axes (cell types) rather than concentrated in a few. Figure 12 shows the full raw spectrum against this bulk edge directly.

Table 8: Top-6 spiked eigenvalues (of 33 total), single-cell TNBC, patient CID44971, against the Marchenko–Pastur edge $\lambda_+ = 2.055$.

Rank	Eigenvalue	$/\lambda_+$	Top-5 loading genes
1	110.30	53.7×	B2M, TMSB4X, HLA-B, CORO1A, CD3E
2	61.21	29.8×	GAPDH, PFN1, FOS, CFL1, IGFBP4
3	43.79	21.3×	RPS12, RPL18A, CST3, LGALS1, RPS27
4	27.13	13.2×	TYROBP, FCER1G, CD68, C1QB, HLA-DRA
5	22.03	10.7×	RPL3, GNB2L1, RPS4X, RPS14, RPL29
6	17.50	8.5×	AZGP1, TACSTD2, TM4SF1, CLDN4, CXCL2

Table 8 gives an independent, real check on what these spikes actually are, using the sample's own cell-type annotation (never given to the RMT method, which sees only the expression matrix): rank 1 (B2M, CD3E, PTPRC among its top loadings) is a pan-immune/T-cell axis, matching this sample's real composition (T-cells are its single largest population, 4366 of 7986 cells); rank 4 (TYROBP, FCER1G, CD68, C1QB, HLA-DRA) is a myeloid/macrophage axis, matching the 684 Myeloid cells present; rank 6 (TACSTD2, CLDN4, TM4SF1) is an epithelial axis, matching the 894 Cancer Epithelial + 735 Normal Epithelial cells present. Ranks 2, 3, and 5, by contrast, load almost entirely on ribosomal-protein genes (RPS/RPL) and stress/ dissociation-response genes (FOS) — a well-documented technical confound in single-cell RNA-seq (ribosomal content tracks cell size and sequencing depth more than cell identity, and FOS induction is a known tissue-dissociation artifact), not a novel biological program, and we report them as such rather than interpreting them as coexpressed biology. Real spikes and known technical-artifact spikes sit side by side in the same spectrum here, distinguishable only by checking their loadings against independent metadata — a caveat the bulk cohorts' own spike interpretation (Sec. 2.2) did not need to confront as sharply, since bulk RNA-seq has no equivalent per-cell dissociation-stress axis.

The tau-sweep threshold selection (Sec. 2.4's method, unchanged) selects $\tau^* = 0.10$ (Figure 11), substantially sparser than either bulk cohort's threshold (0.45, 0.25), consistent with single-cell gene-gene correlations being weaker than bulk patient-level correlations (dropout noise attenuates them). A Monte Carlo permutation null (identical recipe to Sec. 2.5, 20 replicates, `scripts/24_singlecell_monte_carlo_null.py`) confirms this is real structure, not a dense-matrix artifact: every one of 20 permutations selects exactly $\tau = 0.035$ (zero variance across seeds), well below the real $\tau^* = 0.10$.

Two honest caveats on this specific number, both surfaced only by looking directly at the unfolding rather than trusting the sweep table alone. First, at τ^* itself the two KS statistics are close (Poisson 0.139 vs. GOE 0.133, Figure 11 left panel) — a thinner margin than the bulk cohorts show at their own τ^* (Table 9), even though the sweep's overall shape is unambiguous (a wide, clearly-GOE region at $\tau \leq 0.075$ giving way to a wide, clearly-Poisson region at $\tau \geq 0.125$, with 0.10 sitting exactly at the crossover rather than deep inside either regime). Second, the degree-7 polynomial unfolding used throughout this paper (Sec. 1.1) is a global fit calibrated on the bulk cohorts' much sparser thresholded networks (1.3% and 13.0% edge density at their own τ^*); at $\tau^* = 0.10$ here the thresholded network is denser still (20.4%), and 4% of the unfolded spacings land above $s = 4$ (some far above, reflecting a poorly-conditioned tail fit rather than genuine large gaps), excluded from Figure 11 and disclosed here rather than silently dropped. The visible histogram itself reads as a marginal case, not a clean GOE match: it exceeds even the Poisson curve's height near $s = 0$, more consistent with the thin KS margin above than with a textbook Wigner-surmise fit. Neither caveat overturns the sweep's central finding (a real, permutation-confirmed, sparser-than-bulk threshold), but both are stated so the result is not read as cleaner than it is; a local or adaptive unfolding scheme, rather than one global polynomial shared with the much sparser bulk cohorts, is the concrete fix for a future round.

Read together, this is Line 4 run on real data for the first time in this program, and it reproduces the bulk cohorts' central finding (real, permutation-confirmed RMT threshold structure, spiked eigenvalues beyond the Marchenko–Pastur edge) in a completely different regime ($q \ll 1$ instead of $q \gg 1$), while also surfacing a genuine single-cell-specific caveat the bulk data never raised (technical dissociation/ribosomal spikes indistinguishable from biological ones without independent metadata). It also directly resolves the open question Sec. 2.8 left about single-cell pseudo-time: this sample's own (whole-sample) PC1 does not provide the stronger ordering axis that line was hoping for, so that revisit needs genuine single-lineage trajectory inference rather than single-cell data of any kind — checked directly, not just proposed, immediately below. One patient, one subtype, is the honest scope of this result; extending it across the other 8 TNBC patients in the series (pooled, with an explicit batch-effect check this single-patient analysis did not need) is the direct next step.

2.15 Single-lineage pseudo-time: a corrected retest, still null, with a clarifying result

The corrected candidate this Roadmap item itself proposed — pseudo- time restricted to a single, continuously varying lineage, rather than the whole 9-cell-type mixture — is directly available in the same sample: 894 Cancer Epithelial cells (Sec. 2.14), the largest non-immune population and the natural candidate for a proliferation or differentiation gradient. Running the identical PC1-ordering, Dyson-trajectory, and wavelet-decomposition pipeline used elsewhere in this paper (Sec. 6.7) on this subset alone, rather than assuming the restriction helps, gives three specific, honest answers.

First, the ordering-axis hypothesis itself does not survive the check: PC1 restricted to the 894 Cancer Epithelial cells explains 8.6% of variance — *not* an improvement over the whole-sample single-cell PC1 (9.7%, Sec. 2.8) or bulk-patient PC1 (10.2%, Sec. 2.8), if anything slightly worse. Restricting to a single lineage removes the categorical T-cell-vs-everything-else split that was diluting the whole-sample PC1, but it does not, on its own, turn this population into the strong continuous gradient the Roadmap anticipated; whatever developmental or proliferative structure this lineage has is evidently not concentrated in a single dominant linear direction either.

Second, the wavelet retest itself (Line 5) remains a genuine null on this corrected axis, exactly as it was on the uncorrected one (Sec. 2.8): the coarse and fine coexpression networks share 0 of their top-10 hub genes, and a 20-replicate random-cell-order control shows this bare result is not even unusual — every one of the 20 random orderings matches or exceeds the real ordering's 0/10 overlap. The specific genes involved are nonetheless thematically sharp and worth recording: the coarse-scale network's hubs are proteasome/ ribosome-biogenesis genes (*PSMA7*, *PSMD8*, *ADRM1*); the fine-scale network's hubs are almost entirely mitochondrially encoded respiratory-chain genes (*MT-ND4L*, *MT-CO1*, *MT-ND1*, *MT-CO2*, *MT-ND3*, *MT-CO3*, *MT-ATP6*, *MT-ND4*). Two biologically coherent, mutually exclusive gene programs sit at the two scales, but the coarse/fine *split itself* is statistically indistinguishable from an arbitrary cell ordering — a real, specific null, not merely an absence of structure, and the same caveat applies here as in Sec. 2.8: this is a data/ordering problem, not evidence against the wavelet method, whose own positive control (Sec. 2.8) already showed it works correctly given a strong enough ordering axis.

Third, and the most useful methodological clarification this retest provides: the Dyson-trajectory diagnostic itself (Line 4(ii)) does show the qualitative approach-and-reopen pattern along this pseudo-time for most pairs (λ_4–λ_5 through λ_7–λ_8), but a direct comparison against 20 independent *random* cell orderings (not attempted for the bulk cohorts' original patient-order scan, only their reproducibility *across* random orders) shows random orderings reproduce the same repulsion pattern at comparable or higher rates for every one of those pairs (e.g. λ_6–λ_7: real order shows clear repulsion, and so do 18/20 random orders). This does not contradict the bulk-cohort Dyson result: it clarifies what that result, and this paper's own framing of Dyson's SDE (Definition 4), already say — "time" in this diagnostic is sample size accumulating, not a specific biologically meaningful axis, so the repulsion signature is expected to appear under *any* growing sequence of real correlation matrices built from the same underlying data, ordering included. Put plainly: the bulk-cohort multi-order reproducibility check (Sec. 2.9, Fig. 6) already tested exactly this, at scale, months before this specific single-cell comparison was run, and got the same answer; this retest confirms it again, explicitly, on an entirely different dataset and cell population, rather than leaving it as an assumption carried over from the bulk result. Figure 13 shows the comparison directly.

The corrected candidate above was still a linear one (PC1, only the population it was computed on changed); the sharper version of the same retest replaces PC1 itself with a genuinely non-linear estimator built for exactly this purpose. Diffusion pseudotime (DPT, 15) builds a cell-cell diffusion process from a k-nearest-neighbor similarity graph rather than projecting onto one global linear axis, and orders cells by their diffusion distance from an extremal root cell; we implement it directly (no external single-cell package was available in this environment) and run it on the same 894 Cancer Epithelial cells (Sec. 6.8). This ordering is not a relabeling of PC1 under another name: the two rankings correlate at Spearman $\rho = -0.481$ ($p = 6.5 \times 10^{-53}$) — a real, significant association (diffusion pseudotime and a linear variance-maximizing axis are not independent processes on the same data) but far from the near-± 1 that would mean DPT is just PC1 relabeled.

Rerunning the identical Dyson-trajectory scan and its random-order control along DPT gives the same qualitative answer as the PC1 version above, now for a third time on a genuinely different ordering: λ_4–λ_5 through λ_6–λ_7 show clear repulsion under DPT, but 15–18 of 20 random cell orderings show the identical pattern for those same pairs, confirming again that this diagnostic's repulsion signature tracks growing sample size rather than which specific ordering is used — the bulk-cohort check (Sec. 2.9), the PC1-based single-cell check above, and this DPT-based check now agree on the same point across three unrelated datasets and orderings. The wavelet retest, however, gives a cleaner and more useful negative result than either PC1 attempt did: coarse and fine hub genes still share 0 of 10, matched or exceeded by all 20 random-order replicates, exactly as with global PC1 (Sec. 2.8) and lineage-restricted PC1 (above). Three independent ordering axes on this same cell population — one categorical-mixture PC1, one single-lineage PC1, and now one genuinely non-linear diffusion pseudotime — all fail the identical wavelet test identically. That consistency itself is informative: it is no longer plausible to blame the wavelet null on a poorly chosen ordering axis, of any kind tried so far; whatever is missing is more specific than "the ordering wasn't good enough," and a fair retest now needs either a substantially larger single-cell cohort (more cells per lineage than this one sample provides) or an experimentally measured time axis (e.g. a time-course or RNA-velocity-derived rate, not any expression- derived ordering) before trying again.

2.16 3D genomic structure: Line 3 attacked with real Hi-C data

Line 3 was, until this round, the one originally proposed line never attempted with real data at all — earlier rounds of this paper reported no public, patient-cohort-scale TNBC Hi-C dataset as identified, a claim already corrected once the group's own Reyes-Gopar et al. [31] was found (Sec. 3, Roadmap). We now run the identical RMT threshold/spiked-eigenvalue machinery used throughout this paper on real Hi-C data for the first time: GSM5098082 ("TNBC_Tissue3"), the smallest of GSE167150's three real TNBC tumor `.hic` files, chromosome 18 at 100kb resolution (Sec. 6.13). One chromosome, the smallest of the three real TNBC tumors, is a deliberately narrow first attempt, matching the concretely-sized single-file, single-chromosome first step this Roadmap item already specified.

A raw Hi-C contact matrix is not directly comparable to a gene coexpression matrix: nearby genomic bins always contact each other more than distant ones regardless of any real biology, a genomic-distance trend with no analogue in expression data. Following Lieberman-Aiden et al. [21], we therefore first divide every diagonal band of the raw contact matrix by its own genome-wide mean (observed over expected, O/E), removing that distance trend, and then compute the Pearson correlation matrix of the O/E matrix's rows — exactly the construction Lieberman-Aiden et al. [21] introduced to reveal A/B chromatin compartments, and the direct structural analogue of the gene-gene correlation matrices used everywhere else in this paper. 31 of chr18's 781 100kb bins have zero total contact coverage (unmappable or centromeric regions, a known, purely technical Hi-C artifact); these are dropped before O/E normalization and correlation, the direct genomic-bin analogue of this paper's own single-cell gene filter (Sec. 6.12), leaving 750 bins.

The identical tau-sweep/NNSD method (Sec. 2.4) selects $\tau^* = 0.15$ on this bin-bin correlation matrix (Figure 14, center panel). A Monte Carlo permutation null — each bin's own O/E profile independently shuffled across the other bins, 20 replicates, identical recipe to Sec. 2.5 — finds a transition at $\tau = 0.10$ or no clean transition at all in 20/20 replicates, never at 0.15 or above: the real threshold is not a dense-matrix artifact, the same qualitative confirmation the bulk cohorts and the single-cell sample both received. The bin-bin correlation matrix itself (Figure 14, left panel) shows the plaid/checkerboard pattern that is the visual signature of A/B compartments in every published Hi-C compartment analysis since Lieberman-Aiden et al. [21], and the top eigenvector's sign pattern splits the 750 retained bins into contiguous genomic blocks (457 positive, 293 negative bins, Figure 14, right panel), not an unstructured scatter of signs.

A plaid pattern and a contiguous sign split are visually suggestive of compartments, but neither is proof: this round checks the interpretation directly, against two real, independent genomic features the RMT method itself never sees, computed from the hg19 reference sequence and the hg19 RefSeq gene table (the genome build the `.hic` file's own header declares, read before downloading anything — Sec. 6.14). The compartment interpretation holds up: the top eigenvector's sign correlates with real GC content ($r = 0.452$, $p = 4.3 \times 10^{-39}$) and with real RefSeq gene density ($r = 0.306$, $p = 9.3 \times 10^{-18}$) across the 750 retained bins (Figure 15, left panel), and the two eigenvector- defined groups differ sharply on both features directly (positive-sign bins: mean GC 0.409, 5.32 transcripts/bin; negative-sign bins: mean GC 0.381, 1.61 transcripts/bin; Welch $t = 10.84$, $p = 1.8 \times 10^{-25}$ for GC and $t = 7.22$, $p = 1.5 \times 10^{-12}$ for gene density) — higher GC, gene-dense bins on one side, lower GC, gene-poor bins on the other, exactly the A/B compartment signature Lieberman-Aiden et al. [21] first reported, not merely a resemblance to it.

One chromosome, one tumor cannot separate a real biological signal from one sample's own idiosyncrasy, so we ran the identical pipeline — O/E normalization, RMT threshold sweep, permutation null, GC/gene- density validation, no changes — on the other two real TNBC tumors in GSE167150 and the matched-normal file (`scripts/33_hic_multi_sample.py`). All four samples give a real, permutation-confirmed threshold well above the null ceiling of $\tau = 0.10$ ($\tau^* = 0.25$, 0.25, 0.15, 0.40 for TNBC_Tissue1/2/3 and Normal_Tissue respectively), and all four show the same GC/gene-density compartment signature independently ($r_{\mathrm{GC}} = 0.63$, 0.67, 0.45, 0.60; all $p < 10^{-38}$) — the compartment finding is not one sample's coincidence.

What is not simply a clean tumor-vs-normal story is the compartment *map* itself: pairwise sign agreement between samples (Figure 15, right panel) shows TNBC_Tissue1, TNBC_Tissue2, and the normal sample mutually agreeing on 87–92% of bins, while TNBC_Tissue3 — the sample examined first, above — agrees with each of the other three on only 74–76% of bins, including the normal sample. If compartment structure tracked malignancy status cleanly, the three tumors should cluster together against the one normal sample; instead, one tumor (Tissue3) is the outlier against all three others, tumor and normal alike. With only three tumors this cannot distinguish real inter-patient heterogeneity — well documented for TNBC generally [11] — from a technical difference specific to one sample (sequencing depth, tissue quality), but it does mean the natural first hypothesis (compartments differ systematically between TNBC and normal breast tissue on chr18) is not what these four real samples show; a larger cohort is needed before either explanation can be preferred over the other.

This is still a single-chromosome result, not a validated genome-wide compartment map, but it now stands on more than visual resemblance: the identical spectral machinery this paper uses for gene expression applies, without modification beyond the one standard O/E preprocessing step, to real Hi-C contact data; the resulting threshold is real by the same permutation-null standard applied everywhere else in this paper, in all four real samples tested; and the compartment reading of the top eigenvector is now independently confirmed against real genomic annotation, also in all four samples. Extending beyond chr18 to the rest of the genome, and resolving whether TNBC_Tissue3's divergence reflects biology or a technical artifact with more tumors, are the concrete next steps, stated in Sec. 4.

3 Discussion

Four of the six original program lines were run on real data in this paper and stand as positive results: RMT coexpression thresholding, validated by a Monte Carlo permutation null rather than the asymptotic Marchenko–Pastur/Wigner-surmise comparison alone; spectral denoising/spiked eigenvalues, further validated by spectral entropy and a Tracy–Widom edge-scaling check; a first Dyson-repulsion diagnostic; and Line 2's large-scale network characterization, confirming a small-world/modular signature beyond what either random-graph null produces. Line 5 (wavelet multi-scale decomposition), the ML recognition benchmark, and now Line 6b (Sec. 2.11) all came back with real, nuanced negative or mixed results, reported as such rather than dropped: the wavelet method's real-data application failed a random-ordering control, but a synthetic positive control then confirmed the method itself works correctly at both scales given enough signal, narrowing the real-data failure down to a data problem (a weak 10.2%-of-variance ordering axis), not a method problem; TNBC-vs-LumA classification is close to a ceiling that a genetic-algorithm search can nudge but not meaningfully break, so it does not discriminate feature-selection methods either way; and the classifier's own weight-matrix eigenvalues do not repel as reliably as the data's do over a full training run, but the repulsion that is present concentrates almost entirely in the pre-convergence transient, a real, quantified pattern rather than an absence of one. Single-cell resolution (Line 4), the other line previously not attempted at all, has since been run for real (Sec. 2.14): GSE176078 [38], one TNBC patient (7986 cells), the identical RMT machinery, in a regime where $q = p/n$ inverts relative to the bulk cohorts exactly as anticipated, with a real permutation- confirmed threshold and a new, single-cell-specific caveat (ribosomal/ dissociation-stress spikes indistinguishable from biological ones without independent cell-type metadata) that the bulk analysis never had to face. 3D genomic structure (Line 3, Hi-C), the line left genuinely unattempted through several earlier rounds, has now been run too (Sec. 2.16): GSE167150 (TNBC Hi-C with matched contralateral healthy tissue, 31, found only after an earlier round of this paper reported no such Hi-C dataset as identified), one chromosome (chr18) of the smallest of its three real TNBC tumor `.hic` files, observed/expected normalization following Lieberman-Aiden et al. [21], and the identical RMT threshold machinery — a real, permutation-confirmed threshold and a bin-bin correlation matrix with the plaid pattern characteristic of A/B chromatin compartments, though that compartment reading is not yet independently validated against genomic annotation. All six originally proposed lines have now been run on real data at least once; what remains, for each, is depth (more chromosomes and tumors for Line 3, more patients for Line 4, other gene subsets for the Dyson-repulsion lines, a rescaled detector for Line 6b) rather than a first attempt.

The Dyson-repulsion result (Sec. 2.9) was, in an earlier round, the newest and least tested of the four executed lines, resting on one cohort, one 60-gene subset, and one random patient-inclusion order; the multi-order reproducibility check (Fig. 6) closed that specific gap and resolved the one open question it had left (whether λ_2–λ_3 repels at all) in the affirmative. Corollary 1, from that same round, gives the one lingering data-side anomaly (λ_1–λ_2's unreliability) a quantitative account directly from the same SDE. What had remained open — one cohort only — is now closed too: the identical check on LumA (Sec. 2.10) replicates the pattern in a cohort more than three times larger, with the same qualitative signature (the pair(s) adjacent to the largest, most spike-like eigenvalues the least reliable), and Corollary 1's account extends naturally to LumA's two unreliable pairs given its larger spiked-eigenvalue count (Table 2). What remains open now is narrower still: extending the multi-order check to gene subsets other than the top-60-by-variance choice used throughout.

The single-cell side of the same question (Line 4(ii), Sec. 2.15) gives a different kind of answer, now checked twice over. Restricting pseudo-time to the 894-cell Cancer Epithelial lineage does not improve the ordering axis over the whole-sample PC1 (8.6% vs. 9.7% variance); replacing that linear axis with diffusion pseudotime [15], a real non-linear estimator built for exactly this purpose (Spearman $\rho = -0.481$ against PC1 — a genuinely different ordering, not PC1 under another name), changes nothing about either conclusion: the wavelet retest stays a null indistinguishable from a random ordering for a third distinct axis in a row, and the Dyson-repulsion signature at single-cell resolution again tracks growing sample size rather than which ordering is used (random cell orders match or exceed it). This is a clarification of what "time" means in this diagnostic (Definition 4), consistent with, not contradicted by, the bulk-cohort reproducibility results, and three-for-three across unrelated orderings now closes the "was it the ordering?" question for the wavelet line specifically: it was not, so the concrete open item is no longer any expression-derived ordering, linear or not, but an experimentally measured time axis or a substantially larger single-cell cohort.

Line 6b's classifier-side extension (Sec. 2.11) is now run rather than merely proposed, and closes with a specific, mechanistic finding rather than a flat null: weight-matrix repulsion is real but confined to the pre-convergence training transient, a 12-fold difference between the early and late windows (`scripts/22_classifier_dyson_early_late.py`). The direct follow-up test this result proposed for itself — does a longer genuine transient produce a longer detectable-repulsion window? — has also now been run (Sec. 2.12): at a 5× smaller learning rate the transient lengthens by $\sim 11\times$ (not the 5× a linear relationship would predict), the post-convergence null gets even cleaner (0/140 vs. 3/140), and the qualitative claim holds, but the raw pre-convergence detection count actually falls (11/140 vs. 36/140) because the closest-approach/reopening criterion's fixed 5-epoch window does not rescale with the slower per-epoch dynamics — a real limitation of the detector, honestly surfaced rather than smoothed over, and a concrete methodological fix (a window sized to the empirical per-epoch step, not a fixed epoch count) for a future round.

The classification ceiling effect in Table 7 is itself an argument for redirecting the ML line toward harder, clinically framed tasks rather than toward a different feature-selection method on the same easy task: TNBC-vs-LumA is close to solved by construction, and no amount of better feature engineering will show a visible gap on it.

All numbers reported above come directly from the scripts in `scripts/`, run against the two real cohorts in `data/TCGA-BRCA/` and the real single-cell sample in `data/GSE176078_singlecell/`; none are estimated, recalled, or adjusted after the fact. Re-running `scripts/01_build_cohorts.py` through `scripts/25_singlecell_figures.py` in order (see "Code and reproducibility" in Methods for the named exceptions) reproduces every number and figure in this paper from the raw downloaded files.

4 Future directions

All six originally proposed lines, including Line 3 and Line 4, both previously unattempted with any real data, and Line 6b, previously an untried extension, have now been run at least once (Sec. 2.16, 2.14, 2.11); every entry below is kept, relabeled "revisited," to state the sharper next step each result now points to rather than the original, now-resolved question of whether it could be attempted at all.

4.1 Line 3 revisited: beyond one chromosome

This line was reported as blocked, in an earlier round, for lack of a public patient-cohort-scale TNBC Hi-C dataset; that claim was corrected once Reyes-Gopar et al. [31] — Hernández-Lemus's own group — was found (GEO accession GSE167150, three real TNBC tumor `.hic` files and a matched-normal file among six 40 kb–500 kb-resolution files per condition, 270MB–2.5GB each, 6.2GB total for the series), and every concrete next step that correction proposed has now been run for real (Sec. 2.16): chr18 of all three real TNBC tumors and the normal sample, observed/expected normalization [21], the identical RMT threshold machinery, and independent validation of the compartment interpretation against real GC content and RefSeq gene density. All four samples give a real, permutation-confirmed threshold and the predicted GC/gene-density compartment signature; what they do *not* give is a clean tumor-vs-normal split (TNBC_Tissue1, TNBC_Tissue2, and the normal sample mutually agree on 87–92% of compartment calls, TNBC_Tissue3 agrees with each of the other three, tumor or normal, on only 74–76%). The direct next steps this result itself points to: (i) extend from one chromosome to the rest of the genome, now that the per-chromosome pipeline and its validation are both established; (ii) resolve TNBC_Tissue3's divergence with more tumors — three samples cannot distinguish real inter-patient heterogeneity, well documented for TNBC generally [11], from a technical artifact specific to that one sample (sequencing depth, tissue quality); (iii) compare the RMT-selected hub bins directly against Reyes-Gopar et al. [31]'s own network measures (Jaccard similarity between normal and TNBC networks, degree and Z-weighted degree), the comparison those measures were themselves designed for and which this round did not yet attempt.

4.2 Line 5 revisited: three orderings, one answer

The synthetic positive control (Results) already confirms the coarse/fine wavelet pipeline works correctly given enough real signal, so what remains is not a method fix but a data fix. This was checked directly, not just proposed, in three rounds now, on three unrelated ordering axes. First: the global PC1 of the single-cell data (Sec. 2.14) carries only 9.7% of variance, no improvement over bulk-patient PC1's 10.2% (Sec. 2.8), because this sample's real structure is spread across many distinct cell-type axes rather than one dominant gradient. Second, pseudo-time restricted to the 894-cell Cancer Epithelial lineage alone (Sec. 2.15) does not fix the problem either: its own PC1 explains only 8.6% of variance, no better than the whole-sample figure. Third, diffusion pseudotime [15] on that same lineage — a genuinely non-linear estimator, confirmed distinct from PC1 by their -0.481 Spearman correlation, not a restriction of the same linear approach at all — gives the identical wavelet null a third time (Sec. 2.15). Three different orderings failing identically is itself the finding: it rules out ordering choice as the explanation, linear or not, restricted or not. What remains is an experimentally measured time axis — RNA-velocity-style splicing information (not available in this processed count matrix and would require re-processing from raw reads) or a genuine time-course — or enough additional cells, within one lineage, that any reasonable ordering estimator would agree on the same trajectory; a fourth expression-derived pseudo-time estimator on this same 894-cell sample is not a promising next step on its own.

4.3 Line 1 extension: multi-omic spectral denoising

Sec. 2.2's spiked-eigenvalue analysis was run on RNA-seq alone. GSE171958 [6] pairs DNA methylation, RNA-seq, and proteomics on the same TNBC cell lines and is public; the direct extension is a joint spiked-eigenvalue analysis across omic layers (a block correlation matrix with one block per layer), asking whether any spike loads across layers rather than within a single one — a real, checkable version of the "multi-omic denoising" claim in Line 1 of the original proposal, not attempted here since GSE171958 covers only three cell lines, too few for the patient-cohort framing used throughout this paper. Hernández-Lemus & Ochoa [16] survey the statistical, machine-learning, and network-based methods actually used for this kind of integration in cancer more broadly, including the variable-selection and normalization concerns (LASSO-type penalization, platform-specific scaling) directly relevant to the block-correlation extension proposed above; this review, by one of this paper's own co-authors, is the natural methodological reference for that next step rather than a general literature search.

4.4 Line 4 revisited: beyond one patient

Line 4 has now been run on real data (Sec. 2.14): GSE176078 [38], one TNBC patient (CID44971, 7986 cells), the identical RMT threshold/spiked-eigenvalue machinery, a real permutation-confirmed threshold, and an honest new caveat (technical ribosomal/dissociation-stress spikes indistinguishable from biological ones without independent metadata) that the bulk cohorts never had to confront. The direct next steps this result itself points to: (i) extend from one patient to the other 8 TNBC-labeled samples in the same series, pooled, with an explicit check for cross-patient batch effects in the spike structure (the single-patient design here was chosen specifically to avoid that confound, not to avoid the question entirely); (ii) the Dyson-trajectory diagnostic and the wavelet retest have both now been run (Sec. 2.15) on the 894 Cancer Epithelial cells in this same sample, under three different orderings (whole-sample PC1, lineage-restricted PC1, and diffusion pseudotime, 15): none improves the wavelet result, and the Dyson-repulsion pattern proved attributable to growing sample size under all three, not to which ordering is used (a real-vs-random- order comparison not previously run at single-cell resolution). What remains open on this front is therefore not a fourth expression-derived ordering but a genuinely different *kind* of signal — real experimentally measured time (a time-course, or RNA-velocity-style splicing information) — since PC1 (twice) and a real non-linear diffusion estimator have now all been checked and found insufficient. (iii) The global polynomial unfolding used for the NNSD test itself shows a heavier outlier tail at this sample's much denser threshold (20.4% edge density) than the sparser bulk cohorts ever exercise it at (Sec. 2.14); a local or adaptive unfolding scheme, standard in denser RMT applications, is a concrete, self-contained methodological fix worth making before pooling more single-cell samples through the same pipeline.

4.5 Line 6b revisited: beyond one small classifier

Line 6b, previously proposed here as an untried extension, has now been run on real data (Sec. 2.11): a small MLP's weight-matrix Gram eigenvalues do not repel as reliably as the data-side correlation-matrix eigenvalues over a full 300-epoch training run, but the repulsion that is present concentrates almost entirely in the pre-convergence transient (epochs 0–49), a 12-fold difference against the post-convergence window. The direct follow-up this result proposed for itself — does a longer genuine transient produce a longer detectable-repulsion window? — has also now been run (Sec. 2.12, a $5\times$ smaller learning rate): the qualitative claim holds and the post-convergence null gets even cleaner, but the raw detection count itself is not a reliable measure once the underlying timescale changes, because the fixed 5-epoch reopening window used throughout this paper does not rescale with a slower learning rate. The concrete next step this result now points to is narrower and purely methodological: redefine the reopening criterion in units of the empirical per-epoch weight-change magnitude rather than a fixed epoch count, then rerun this same comparison so the early-window counts across learning-rate regimes are directly comparable; only after that fix would a deeper/wider network or a harder, non-plateauing task (the within-TNBC or survival tasks of Sec. 3) actually tell us anything new here, since either change would face the identical rescaling problem on its own.

4.6 RMT threshold vs. ARACNE on the same cohort

The comparison sketched in the Introduction — RMT spectral thresholding against the group's own bootstrap-calibrated ARACNE/NIMEA pipeline — can be run directly on the TNBC cohort already built in `data/TCGA-BRCA/tnbc_expr.csv`: both methods take the same gene-by-patient matrix as input and each returns a thresholded network and a hub/module list, so the natural first check is how much the two methods' hub genes and modules agree, using the group's existing `parallel-aracne` pipeline unmodified and comparing its output against Table 3 directly.

5 Conclusion

Combining a static RMT correlation threshold, a Monte Carlo permutation null, and a Dyson-repulsion eigenvalue diagnostic into one pipeline surfaces real, non-trivial structure in TCGA-BRCA: a sample-size-robust difference in coexpression-network density between TNBC and luminal A that a permutation null confirms is not a threshold artifact, biologically coherent (collagen/ECM, interferon, keratin) spiked correlation components identified by a Marchenko–Pastur threshold and not by eigenvector localization alone, and level repulsion among coexpression-matrix eigenvalues as sample size grows, checked for reproducibility across 20 random patient orders in *two* real cohorts (TNBC and LumA), with a quantitative corollary (Corollary 1) explaining directly, from the same SDE, which pairs fail to repel reliably and by how much in both cohorts alike.

The same discipline surfaces three honest negative results, reported as such instead of dropped. TNBC-vs-LumA classification is too easy a benchmark to distinguish feature-selection methods. A wavelet multiscale decomposition produced a headline result statistically indistinguishable from a meaningless random ordering, checked now on four unrelated orderings across two data types (bulk-patient PC1, whole-sample and lineage-restricted single-cell PC1, and single-cell diffusion pseudotime) and failing identically every time — evidence strong enough to rule out ordering choice as the explanation, not just suggest it. And the classifier's own weight-matrix eigenvalues, tracked across SGD training at two different learning rates, repel far less reliably than the data's do over a full run; what repulsion exists concentrates almost entirely in the pre-convergence transient regardless of how long that transient is made, a specific, quantified pattern rather than an unexplained absence, with one further honest complication surfaced by the second learning-rate regime: the fixed-epoch-window detector used throughout this paper does not itself rescale with training speed, so its raw counts should not be compared directly across learning rates without that fix.

The same pipeline runs, for the first time in this program, on two further data modalities beyond bulk expression. On single-cell RNA-seq (Line 4), it gives a real, permutation-confirmed threshold in a $q \ll 1$ regime the bulk cohorts cannot reach, with spikes that split cleanly into real cell-type axes (immune, myeloid, epithelial) and known technical artifacts (ribosomal content, dissociation stress) once checked against independent metadata. On Hi-C contact data (Line 3), the one line left completely unattempted through several earlier rounds, it gives a real, permutation-confirmed threshold and a bin-bin correlation matrix whose top eigenvector correlates with real GC content and gene density exactly as the A/B compartment literature predicts — not asserted from a plaid-looking heatmap, but confirmed against independent genomic annotation, and replicated across all three real TNBC tumors and the matched normal sample in GSE167150, where one tumor (Tissue3) turns out to be the outlier against the other three rather than tumor and normal cleanly separating.

Every one of the six originally proposed research lines has now been run on real data at least once, each with either a positive result, an honest null, or both. That is the actual shape of this paper: not six confirmations, but six real attempts, reported with whatever they turned up. A validation-first collaboration should want exactly this — the real results, the honest nulls, and the negative checks on the program's own prior assumptions, all in one place. What is left is depth, not a first attempt: extending the two-cohort Dyson-repulsion scan to gene subsets beyond the top-60-by-variance choice used throughout; rescaling the classifier-repulsion detector to the empirical per-epoch step size before comparing learning-rate regimes directly; pooling the single-cell result across the other 8 TNBC patients in GSE176078 with an explicit batch-effect check, and finding an experimentally measured time axis for single-cell trajectory work, since three expression-derived pseudo-time estimators have now all given the same negative answer; extending the validated chr18 compartment call to the rest of the genome and resolving, with more tumors, whether TNBC_Tissue3's divergence is biology or artifact; and replacing the classification benchmark with a task where a feature-selection advantage, if real, would actually be visible. None of these next steps is a first attempt at an unopened question anymore — every one of them is a sharper version of a question this paper has already answered once, for real, with real data.

6 Methods

6.1 Data

RNA-seq expression (RSEM, $\log_2(x+1)$-normalized, 20530 genes, 1218 samples) and clinical annotation (1247 samples, TCGA Nature 2012 supplement fields) for TCGA-BRCA were downloaded from the UCSC Xena TCGA hub [13, 32] on 4 September 2026 (`data/TCGA-BRCA/HiSeqV2.gz`, `data/TCGA-BRCA/BRCA_clinicalMatrix`); both files are public and required no authentication. TNBC status was defined as `ER_Status_nature2012 == Negative` and `PR_Status_nature2012 == Negative` and `HER2_Final_Status_nature2012 == Negative`; LumA status as `PAM50Call_RNAseq == LumA` [28]. Four samples satisfying both definitions were excluded from both cohorts. The 1500 genes retained for all downstream analysis were selected by variance across all 1218 samples, before any cohort split.

6.2 Spectral entropy and edge-fluctuation scaling

Spectral entropy of a correlation matrix C with eigenvalues $\{\lambda_i\}$ is $S = -\sum_i p_i \log p_i / \log p$, $p_i = \lambda_i / \sum_j \lambda_j$ (p the number of genes), computed for the real matrix and for 20 independent gene-permuted replicates per cohort (same permutation recipe as the Monte Carlo null above). The edge-fluctuation scaling check generates i.i.d. $\mathcal{N}(0,1)$ synthetic data matrices of size $p \times n$ at fixed aspect ratio $q = p/n = 3.0$, for $n \in \{60, 120, 240, 480, 960\}$ ($p = 3n$ accordingly), records the largest eigenvalue of $\mathrm{corr}(X)$ over 60–400 replicates per n (more replicates at smaller, cheaper n), and fits $\log(\mathrm{std})$ vs. $\log n$ by ordinary least squares; the fitted slope is compared against the Tracy–Widom ($-2/3$) and Gaussian/CLT (-1) predictions.

6.3 RMT threshold selection

For a correlation matrix C and threshold τ, we form $A_{ij} = C_{ij}\mathbb{1}[|C_{ij}| \geq \tau]$ for $i \neq j$ and $A_{ii} = 0$, and compute its eigenvalues $\{\lambda_k\}$. The spectrum is unfolded by fitting a degree-7 polynomial to the empirical cumulative distribution of $\{\lambda_k\}$ and rescaling so mean spacing is 1 [22]. Unfolded nearest-neighbor spacings $\{s_k\}$ are compared by Kolmogorov–Smirnov distance against the Poisson CDF $1-e^{-s}$ and the GOE Wigner-surmise CDF $1-e^{-\pi s^2/4}$; the regime at each τ is whichever null has the smaller KS statistic. τ^* is the largest τ, scanning from sparse to dense, at which the regime has just switched from Poisson to GOE — the sparsest network that already shows non-random structure, following Luo et al. [22]. The Marchenko–Pastur bulk edge for a $p \times n$ correlation matrix under the null is $\lambda_\pm = (1 \pm \sqrt{p/n})^2$ [24], reported alongside each cohort for reference but not used directly in threshold selection. Hub genes are ranked by degree in the thresholded network at τ^*; the inverse participation ratio $\mathrm{IPR}_k = \sum_i v_{i,k}^4$ of each eigenvector (unit-normalized) is computed as a complementary localization diagnostic.

6.4 Monte Carlo permutation null

For each cohort, each gene's expression values were independently permuted across patients (shuffling within each row), 20 times, destroying every real gene–gene correlation while keeping each gene's own marginal distribution exactly fixed. The identical RMT threshold sweep (previous subsection) was rerun on each permuted correlation matrix, recording whether a Poisson→GOE transition appeared at all and, if so, at what τ.

6.5 Correlation heatmaps

Gene order for Figure 4 was fixed by average-linkage hierarchical clustering on the distance $d_{ij} = 1 - |C_{ij}|$ (i.e. two genes are "close" if strongly correlated *or* anti-correlated), applied to the raw correlation matrix and reused unchanged for the thresholded panel, so both panels are directly comparable gene-for-gene.

6.6 Dyson eigenvalue trajectories

Restricting to the 60 most variable TNBC genes, patients were placed in one fixed random order (seed 1) and the correlation matrix $C(X_{:,1:n})$ recomputed for every $n = 15, \ldots, 119$; the top 8 eigenvalues of each $C(X_{:,1:n})$ form the trajectories in Figure 5. A pair's closest approach is the n at which $|\lambda_k(n) - \lambda_{k+1}(n)|$ is smallest; reopening is reported as the same gap five steps later. The reproducibility check (Fig. 6) repeats this identical procedure under 20 independent random patient orders (seeds 0–19) and classifies each pair, per order, as clear repulsion if the gap reopens to $\geq 1.5\times$ its minimum on both the 5-steps-before and 5-steps-after sides of the closest approach (clipped at the sample boundary), partial if only one side qualifies, inconclusive if the closest approach is at the boundary $n = 119$ itself (still narrowing, no reopening observable within the data), and none otherwise. `scripts/27_luma_dyson_repulsion.py` repeats this identical procedure, unchanged, on LumA's own top-60 most-variable genes (seed 1 for the single-order scan, seeds 0–19 for the reproducibility check, $n = 15, \ldots, 430$).

6.7 Single-cell pseudo-time: single-lineage retest

The 894 Cancer-Epithelial-annotated cells within the same GSE176078 CID44971 sample (Sec. 6.12) were extracted by real cell-type label (`celltype_major`, never used elsewhere in the pipeline) before any further processing. `scripts/26_singlecell_epithelial_pseudotime.py` then applies, unchanged: (i) the same CP10K/log1p normalization and $\geq$3%-of-cells detection filter as Sec. 6.12, restricted to this 894-cell subset; (ii) PC1 of the top-1500-variance genes within this subset as the pseudo-time ordering, exactly the same construction as the bulk-cohort and whole-sample single-cell PC1 orderings (Sec. 2.8); (iii) the identical Dyson-trajectory scan (previous subsection) on the top-60-variance genes within this subset, $n = 30, \ldots, 894$ cells added along that pseudo-time ordering, plus 20 independent *random* (non-pseudo-time) cell orderings for direct comparison; and (iv) the identical wavelet coarse/fine decomposition (Sec. 2.8 method) on the top-1500-variance genes ordered along the same pseudo-time, plus 20 random-order replicates for the hub-overlap control, following exactly the criterion of script 04b.

6.8 Diffusion pseudotime

Diffusion pseudotime [15] was implemented directly in `numpy/scipy` on the same 894 Cancer Epithelial cells, top-1500-variance genes: a $k = 15$ mutual-nearest-neighbor Gaussian kernel affinity matrix W (bandwidth set by the median squared distance to each cell's own 15th neighbor), symmetric-normalized to $S = D^{-1/2}WD^{-1/2}$ and eigendecomposed directly (real, symmetric, so `eigh` applies, unlike the corresponding non-symmetric Markov transition operator); the transition operator's own eigenvectors are recovered from S's via the standard $D^{-1/2}$ correction, and the top 15 non-trivial components (the leading eigenvalue, ≈ 1, corresponds to the trivial stationary distribution and is dropped) form the diffusion coordinates. The root cell is the one furthest, in diffusion-coordinate space, from the population centroid, a marker-free extremal-point heuristic used because no biological root was specified for this sample; pseudotime is then the closed-form DPT distance from that root, $\mathrm{DPT}(i) = \left(\sum_k [\lambda_k/(1-\lambda_k)]^2(\psi_k(\mathrm{root}) - \psi_k(i))^2\right)^{1/2}$, following Haghverdi et al. [15] exactly. Cells ordered by this distance replace the PC1 ordering of the previous subsection as the input to the identical Dyson-trajectory and wavelet pipelines, with no other change (`scripts/34_singlecell_diffusion_pseudotime.py`).

6.9 Classifier weight-matrix Dyson trajectory

An MLP(20→8→1) (`scikit-learn MLPClassifier`, `tanh` activation, plain SGD, learning rate 0.05, no momentum, batch size 64) was trained on the same 20 RMT-hub genes as Table 3 (standardized), combined 549-patient TNBC+LumA cohort, one epoch at a time (`warm_start=True`, one call to `fit` per epoch) for 300 epochs. After every epoch, the input-to-hidden weight matrix W_1 (20×8) was extracted and its Gram matrix $W_1^\top W_1$ (8×8, real symmetric, positive semi-definite) eigendecomposed; the resulting 8 eigenvalue trajectories across epochs are the direct model-side analogue of the data-side trajectories above, and the identical closest-approach/reopening classification (previous subsection) is applied to them, first for one seed then across 20 independent seeds (initialization and minibatch shuffling, `scripts/19_dyson_classifier_weights.py`, `scripts/20_dyson_classifier_weights_multi_seed.py`). The early/late split (`scripts/22_classifier_dyson_e` reapplies the same classification separately to epochs 0–49 and 50–299 of each of the 20 already-computed trajectories, with no retraining. `scripts/28_classifier_dyson_longer_transient.py` tests directly whether a longer pre-convergence transient produces a longer repulsion window: identical architecture, features, cohort, and 20 seeds, but a 5× smaller learning rate (0.01 vs. 0.05) and a 5× longer run (1500 vs. 300 epochs), the single most direct way to slow convergence without altering anything else that could confound the comparison. Each seed's own empirical convergence epoch (the first epoch after which training accuracy never again drops more than 0.5 points below its own final value) sets that seed's early/late split, rather than reusing the original run's fixed epoch-50 cutoff, which assumed that run's own convergence speed and would beg the question here.

6.10 Machine learning

Logistic regression (`scikit-learn`, 30) with feature standardization, evaluated by 5-fold stratified cross-validation (accuracy and ROC-AUC, random state fixed) on the combined 549-patient TNBC+LumA cohort, for three 20-gene feature sets: a uniformly random draw from the 1500 candidate genes; the top 20 by variance across the combined cohort; and the union of TNBC's and LumA's own RMT hub genes at their respective τ^* (deduplicated, truncated to 20); and a genetic algorithm (`DEAP`, 10) directly searching 20-slot chromosomes of candidate-gene indices, fitness equal to mean 5-fold CV accuracy of the same logistic-regression pipeline on the (deduplicated) gene subset, population 40, 15 generations, tournament selection (size 3), two-point crossover ($p = 0.5$), and per-index mutation ($p = 0.1$ per slot).

6.11 Random-graph network comparison

For each cohort's thresholded network at τ^* (isolated nodes removed), we computed the average clustering coefficient, giant connected component size, and Louvain-community count and modularity score (`networkx`, 14; Louvain rather than greedy-modularity maximization, for tractability on LumA's $\sim$146K-edge network), and compared each against the mean over 3 replicates of two random-graph nulls of matched size: an Erdős–Rényi $G(n, p)$ graph (same node and edge count, no other constraint) and a configuration-model graph built by fast stub-matching (same node count *and* the real network's own degree sequence, testing for structure beyond degree heterogeneity alone).

6.12 Single-cell RMT pipeline

Patient CID44971 (GSM5354531, GSE176078 [38]) was downloaded as a sparse gene $\times$ cell UMI count matrix (29733 genes, 7986 cells) with accompanying cell-type metadata. Counts were normalized per cell to 10^4 total counts (CP10K) and log1p-transformed (standard single-cell practice). Genes detected in fewer than 3% of cells were dropped (10412 of 29733 retained); of these, the top 1500 by variance of the normalized expression (same p and same variance-ranking rule as `scripts/01_build_cohorts.py`) were kept. The Pearson correlation matrix across the 7986 cells, the Marchenko–Pastur edge, the tau-sweep threshold selection, and the Monte Carlo permutation null (20 replicates, gene-wise, identical recipe to Sec. 2.5) all use the exact same code paths as the bulk cohorts (`rmt_threshold_demo.py`), applied to this new data by `scripts/23_singlecell_line4_rmt.py` and `scripts/24_singlecell_monte_carlo_null.py`; the permutation sweep grid was refined to $\tau \in [0.01, 0.085]$ after a diagnostic run showed the permuted null's own transition sits below the bulk cohorts' sweep range (correlations under gene-wise permutation are weaker here, consistent with the smaller real τ^* found for the unpermuted data).

6.13 Hi-C RMT pipeline

GSM5098082 ("TNBC_Tissue3", GSE167150 [31]), the smallest of the series' three real TNBC tumor `.hic` files (645MB), was downloaded with `scripts/download_hic_tnbc_ tissue3.py`, which locates the target member inside GEO's combined `GSE167150_RAW.tar` (6.2GB, mostly unrelated cell-line and normal-tissue files not needed here) via a sequence of small HTTP Range requests against the tar's sequential USTAR headers, then downloads only that one member. The intrachromosomal contact matrix for chromosome 18 at 100kb resolution was extracted with the pure-Python `straw` reader (`pip` package `hic-straw==0.0.6`, the aidenlab/straw project prior to its C++/pybind11 rewrite; no C++ compiler was available in this environment, and this legacy reader works identically for a local file), giving 781 100kb bins. Bins with zero total contact coverage (31 of 781; unmappable/centromeric regions) were dropped, the direct genomic-bin analogue of the single-cell gene-detection filter above, leaving 750 bins. Each diagonal band of the raw contact matrix (all bin pairs at a fixed genomic distance) was divided by its own genome-wide mean (observed over expected, O/E), following Lieberman-Aiden et al. [21], and the Pearson correlation matrix of the O/E matrix's rows was then passed through the identical tau-sweep/NNSD threshold selection (Sec. 2.4) and Monte Carlo permutation null (Sec. 2.5, here permuting each bin's own O/E profile across the other bins, 20 replicates), with no methodological change beyond the O/E step Hi-C data specifically requires (`scripts/30_hic_line3_rmt.py`, `scripts/31_hic_figures.py`).

6.14 Hi-C compartment validation and multi-sample comparison

Independent validation used the hg19 chr18 reference sequence and the hg19 RefSeq gene table (both public, UCSC, downloaded directly for this check), matching the genome build the `.hic` file's own raw header declares (`chrom_hg19.sizes`) rather than an assumed build. GC content per 100kb bin was computed directly from the reference sequence; gene density per bin counted RefSeq transcripts (chr18 only) whose annotated span overlapped that bin. Both were correlated (Pearson) against the top eigenvector's sign/magnitude at the same bins, and compared by Welch's t-test between positive- and negative-sign bins (`scripts/32_hic_compartment_validation.py`). The other two real TNBC tumor `.hic` files (GSM5098079, GSM5098080) and the matched-normal file (GSM5098074), all in GSE167150, were downloaded with the identical targeted-Range-request method as Sec. 6.13 and passed through the identical chr18 pipeline — load, filter zero-coverage bins, O/E normalize, correlate, tau-sweep, permutation null, GC/gene-density validation — with no changes; pairwise compartment-sign agreement between samples was computed only on bins retained (non-zero-coverage) in both samples of each pair, after orienting every sample's top eigenvector sign so that positive means higher GC (avoiding a spurious 0% or 100% agreement score from two samples' eigenvectors having arbitrary, uncorrelated overall sign) (`scripts/33_hic_multi_sample.py`, `scripts/35_hic_validation_figures.py`).

6.15 Code and reproducibility

All analysis code is in `scripts/01_build_cohorts.py` through `scripts/35_hic_validation_figures.py`, numbered in the order they must be run (`04b`, an ordering-control check, `rmt_threshold_demo.py`, the synthetic-data reference implementation, and `download_hic_tnbc_tissue3.py`, a data download helper reused for all four real Hi-C samples, are named descriptively rather than fitting the main sequence; `scripts/21_classifier_dyson_figures.py` plots results from scripts 19–20, `scripts/22` reruns script 20's already-saved trajectories with no retraining, and `scripts/25_singlecell_figures.` plots results from scripts 23–24; `scripts/27_luma_dyson_repulsion.py`, `scripts/26_singlecell_epithelial_ps` and `scripts/34_singlecell_diffusion_pseudotime.py` are independent extensions requiring no output from any other script beyond the raw cohort/single-cell data already described above, `scripts/28_classifier_dyson_longer_transient.py` is an independent rerun of scripts 19–20's training procedure at a different learning rate/epoch budget, `scripts/29_new_figures_ round3.py` plots results from scripts 26–28, `scripts/31_hic_figures.py` plots results from script 30, `scripts/32_hic_compar` and `scripts/33_hic_multi_sample.py` both import functions directly from script 30 rather than duplicating them, and `scripts/35_hic_validation_figures.py` plots results from scripts 32–33); every number and figure in this paper is produced directly by this code from the raw downloaded files (two bulk TCGA-BRCA files, one single-cell sample, four Hi-C samples, one reference-genome sequence, one gene-annotation table), with no intermediate manual editing of results.

Data availability statement

TCGA-BRCA expression and clinical data are public and available from the UCSC Xena TCGA hub (`https://tcga.xenahubs.net`). Single-cell RNA-seq data (patient CID44971) are public and available from GEO accession GSE176078 (GSM5354531). Hi-C data (samples GSM5098079, GSM5098080, GSM5098082, and GSM5098074) are public and available from GEO accession GSE167150. The hg19 reference sequence and RefSeq gene table used for compartment validation are public and available from the UCSC Genome Browser (`http://hgdownload.soe.ucsc.edu`). No proprietary or restricted-access data were used in this paper; all accession numbers needed to reproduce every download are also listed in Methods.

Code availability statement

All analysis code (`scripts/01_build_cohorts.py` through `scripts/35_hic_validation_figures.py`, Methods) is included with this submission as Supporting Information and will be made available in a public code repository without restriction upon publication.

Ethics statement

Not applicable. This study uses only publicly available, de-identified data (TCGA-BRCA, GSE176078, GSE167150); no new human-subjects data were collected and no institutional review was required.

Competing interests

The authors declare no competing interests.

Funding statement

This work received no specific grant from any funding agency in the public, commercial, or not-for-profit sectors.

Author contributions

J.C.G.: Conceptualization, Data curation, Formal analysis, Investigation, Methodology, Software, Validation, Visualization, Writing – original draft. E.H.L.: Conceptualization, Domain expertise, Supervision, Writing – review & editing. Both authors read and approved the final manuscript.

Acknowledgments

We thank the TCGA Research Network and the UCSC Xena platform for public data access, and acknowledge the CSB-IG group's publicly available ARACNE/NIMEA pipeline as a point of methodological comparison (Introduction, Future directions).

References

[1] G. Aarts, O. Hajizadeh, B. Lucini, C. Park, "Dyson Brownian motion and random matrix dynamics of weight matrices during learning," arXiv:2411.13512 (2024). NeurIPS 2024 Workshop on Machine Learning and the Physical Sciences.

[2] A. Altland, M. R. Zirnbauer, "Nonstandard symmetry classes in mesoscopic normal-/superconducting hybrid systems," *Physical Review B* **55**, 1142–1161 (1997).

[3] G. W. Anderson, A. Guionnet, O. Zeitouni, *An Introduction to Random Matrices*, Cambridge Studies in Advanced Mathematics **118**, Cambridge University Press (2010).

[4] J. Baik, G. Ben Arous, S. Péché, "Phase transition of the largest eigenvalue for nonnull complex sample covariance matrices," *Annals of Probability* **33**(5), 1643–1697 (2005).

[5] G. Ben Arous, A. Guionnet, "Large deviations for Wigner's law and Voiculescu's non-commutative entropy," *Probability Theory and Related Fields* **108**, 517–542 (1997).

[6] K. Chappell, K. Manna, C. L. Washam, S. Graw, D. Alkam, M. D. Thompson, et al., "Multi-omics data integration reveals correlated regulatory features of triple negative breast cancer," *Molecular Omics* **17**(5), 677–691 (2021). Data: GEO GSE171958, ProteomeXchange PXD025238. `https://doi.org/10.1039/D1MO00117E`

[7] R. Dorantes-Gilardi, D. García-Cortés, E. Hernández-Lemus, J. Espinal-Enríquez, "k-core genes underpin structural features of breast cancer," *Scientific Reports* **11**, 16284 (2021).

[8] F. J. Dyson, "A Brownian-Motion Model for the Eigenvalues of a Random Matrix," *Journal of Mathematical Physics* **3**, 1191–1198 (1962).

[9] F. J. Dyson, "The Threefold Way. Algebraic Structure of Symmetry Groups and Ensembles in Quantum Mechanics," *Journal of Mathematical Physics* **3**(6), 1199–1215 (1962).

[10] F.-A. Fortin, F.-M. De Rainville, M.-A. Gardner, M. Parizeau, C. Gagné, "DEAP: Evolutionary Algorithms Made Easy," *Journal of Machine Learning Research* **13**, 2171–2175 (2012).

[11] W. D. Foulkes, I. E. Smith, J. S. Reis-Filho, "Triple-Negative Breast Cancer," *New England Journal of Medicine* **363**(20), 1938–1948 (2010).

[12] S. M. Gibson, S. P. Ficklin, S. Isaacson, F. Luo, F. A. Feltus, M. C. Smith, "Massive-Scale Gene Co-Expression Network Construction and Robustness Testing Using Random Matrix Theory," *PLOS ONE* **8**(2), e55871 (2013).

[13] M. J. Goldman, B. Craft, M. Hastie, et al., "Visualizing and interpreting cancer genomics data via the Xena platform," *Nature Biotechnology* **38**, 675–678 (2020).

[14] A. A. Hagberg, D. A. Schult, P. J. Swart, "Exploring Network Structure, Dynamics, and Function using NetworkX," in *Proceedings of the 7th Python in Science Conference (SciPy2008)*, 11–15 (2008).

[15] L. Haghverdi, M. Büttner, F. A. Wolf, F. Buettner, F. J. Theis, "Diffusion pseudotime robustly reconstructs lineage branching," *Nature Methods* **13**(10), 845–848 (2016).

[16] E. Hernández-Lemus, S. Ochoa, "Methods for multi-omic data integration in cancer research," *Frontiers in Genetics* **15**, 1425456 (2024).

[17] A. Kikkawa, "Random Matrix Analysis for Gene Interaction Networks in Cancer Cells," *Scientific Reports* **8**, 10607 (2018).

[18] H. K. Kim, K. H. Park, Y. Kim, S. E. Park, H. S. Lee, S. W. Lim, et al., "Discordance of the PAM50 Intrinsic Subtypes Compared with Immunohistochemistry-Based Surrogate in Breast Cancer Patients: Potential Implication of Genomic Alterations of Discordance," *Cancer Research and Treatment* **51**(2), 737–747 (2019).

[19] R. König, G. Schramm, M. Oswald, et al., "Discovering functional gene expression patterns in the metabolic network of *Escherichia coli* with wavelets transforms," *BMC Bioinformatics* **7**, 119 (2006).

[20] G. R. Lee, R. Gommers, F. Wasilewski, K. Wohlfahrt, A. O'Leary, "PyWavelets: A Python package for wavelet analysis," *Journal of Open Source Software* **4**(36), 1237 (2019).

[21] E. Lieberman-Aiden, N. L. van Berkum, L. Williams, M. Imakaev, T. Ragoczy, A. Telling, et al., "Comprehensive Mapping of Long-Range Interactions Reveals Folding Principles of the Human Genome," *Science* **326**(5950), 289–293 (2009).

[22] F. Luo, Y. Yang, J. Zhong, H. Gao, L. Khan, D. K. Thompson, et al., "Constructing gene co-expression networks and predicting functions of unknown genes by random matrix theory," *BMC Bioinformatics* **8**, 299 (2007).

[23] M. Mahoney, C. Martin, "Traditional and Heavy Tailed Self Regularization in Neural Network Models," *Proceedings of the 36th International Conference on Machine Learning*, PMLR **97**, 4284–4293 (2019).

[24] V. A. Marchenko, L. A. Pastur, "Distribution of eigenvalues for some sets of random matrices," *Mathematics of the USSR-Sbornik* **1**(4), 457–483 (1967).

[25] A. A. Margolin, I. Nemenman, K. Basso, C. Wiggins, G. Stolovitzky, R. Dalla Favera, et al., "ARACNE: An Algorithm for the Reconstruction of Gene Regulatory Networks in a Mammalian Cellular Context," *BMC Bioinformatics* **7**(Suppl 1), S7 (2006).

[26] S. Ochoa, G. de Anda-Jáuregui, E. Hernández-Lemus, "An Information Theoretical Multilayer Network Approach to Breast Cancer Transcriptional Regulation," *Frontiers in Genetics* **12**, 617512 (2021).

[27] S. Ochoa, E. Hernández-Lemus, "Functional impact of multi-omic interactions in breast cancer subtypes," *Frontiers in Genetics* **13**, 1078609 (2022).

[28] J. S. Parker, M. Mullins, M. C. U. Cheang, et al., "Supervised Risk Predictor of Breast Cancer Based on Intrinsic Subtypes," *Journal of Clinical Oncology* **27**(8), 1160–1167 (2009).

[29] F. Passerini, S. Severini, "Quantifying Complexity in Networks: The Von Neumann Entropy," *International Journal of Agent Technologies and Systems* **1**(4), 58–67 (2009).

[30] F. Pedregosa, G. Varoquaux, A. Gramfort, et al., "Scikit-learn: Machine Learning in Python," *Journal of Machine Learning Research* **12**, 2825–2830 (2011).

[31] H. Reyes-Gopar, K. A. Pérez-Fuentes, M. L. Bendall, E. Hernández-Lemus, "Integration of chromosome conformation and gene expression networks reveals regulatory mechanisms in triple negative breast cancer," *Frontiers in Cell and Developmental Biology* **13**, 1597245 (2025). Data: GEO GSE167150.

[32] The Cancer Genome Atlas Network, "Comprehensive molecular portraits of human breast tumours," *Nature* **490**, 61–70 (2012).

[33] T. Tao, V. Vu, "Random matrices: Universality of local eigenvalue statistics," *Acta Mathematica* **206**, 127–204 (2011).

[34] T. Tao, V. Vu, "Random covariance matrices: universality of local statistics of eigenvalues," *Annals of Probability* **40**(3), 1285–1315 (2012).

[35] C. A. Tracy, H. Widom, "Level-spacing distributions and the Airy kernel," *Communications in Mathematical Physics* **159**, 151–174 (1994).

[36] C. A. Tracy, H. Widom, "On orthogonal and symplectic matrix ensembles," *Communications in Mathematical Physics* **177**, 727–754 (1996).

[37] D. Voiculescu, "Limit laws for random matrices and free products," *Inventiones Mathematicae* **104**, 201–220 (1991).

[38] S. Z. Wu, G. Al-Eryani, D. Roden, S. Junankar, K. Harvey, A. Andersson, et al., "A single-cell and spatially resolved atlas of human breast cancers," *Nature Genetics* **53**, 1334–1347 (2021). Data: GEO GSE176078.

A Full threshold sweep

Tables 9 and 10 give the complete sweep underlying Fig. 3 and Sec. 2.4, reproduced verbatim from `results/tnbc_rmt.csv` and `results/luma_rmt.csv`.

Table 9: Full TNBC threshold sweep.

τ	edges	density	KS(Poisson)	KS(GOE)	regime
0.15	383911	0.3415	0.1935	0.1050	GOE
0.20	238751	0.2124	0.2067	0.0949	GOE
0.25	142388	0.1267	0.2191	0.0772	GOE
0.30	82197	0.0731	0.2074	0.0947	GOE
0.35	46347	0.0412	0.2006	0.1272	GOE
0.40	25911	0.0230	0.1986	0.1670	GOE
0.45	**14498**	**0.0129**	**0.2504**	**0.2369**	**GOE (τ^*)**
0.50	7984	0.0071	0.3476	0.3919	Poisson
0.55	4391	0.0039	0.4246	0.5013	Poisson
0.60	2413	0.0021	0.4507	0.5428	Poisson
0.65	1389	0.0012	0.4715	0.5493	Poisson
0.70	814	0.0007	0.5442	0.6313	Poisson
0.75	478	0.0004	0.4712	0.5614	Poisson
0.80	258	0.0002	0.4217	0.4980	Poisson

Table 10: Full LumA threshold sweep.

τ	edges	density	KS(Poisson)	KS(GOE)	regime
0.15	347733	0.3093	0.2008	0.1106	GOE
0.20	223750	0.1990	0.1968	0.1222	GOE
0.25	**146100**	**0.1300**	**0.1899**	**0.1321**	**GOE** (τ^*)
0.30	98072	0.0872	0.1649	0.1653	Poisson
0.35	68007	0.0605	0.1763	0.1845	Poisson
0.40	47940	0.0426	0.1988	0.2305	Poisson
0.45	34153	0.0304	0.2587	0.2875	Poisson
0.50	24361	0.0217	0.3129	0.3706	Poisson
0.55	17034	0.0152	0.3902	0.4708	Poisson
0.60	11560	0.0103	0.4490	0.5313	Poisson
0.65	7427	0.0066	0.5182	0.6094	Poisson
0.70	4493	0.0040	0.5495	0.6333	Poisson
0.75	2553	0.0023	0.5911	0.6804	Poisson
0.80	1389	0.0012	0.4985	0.5881	Poisson

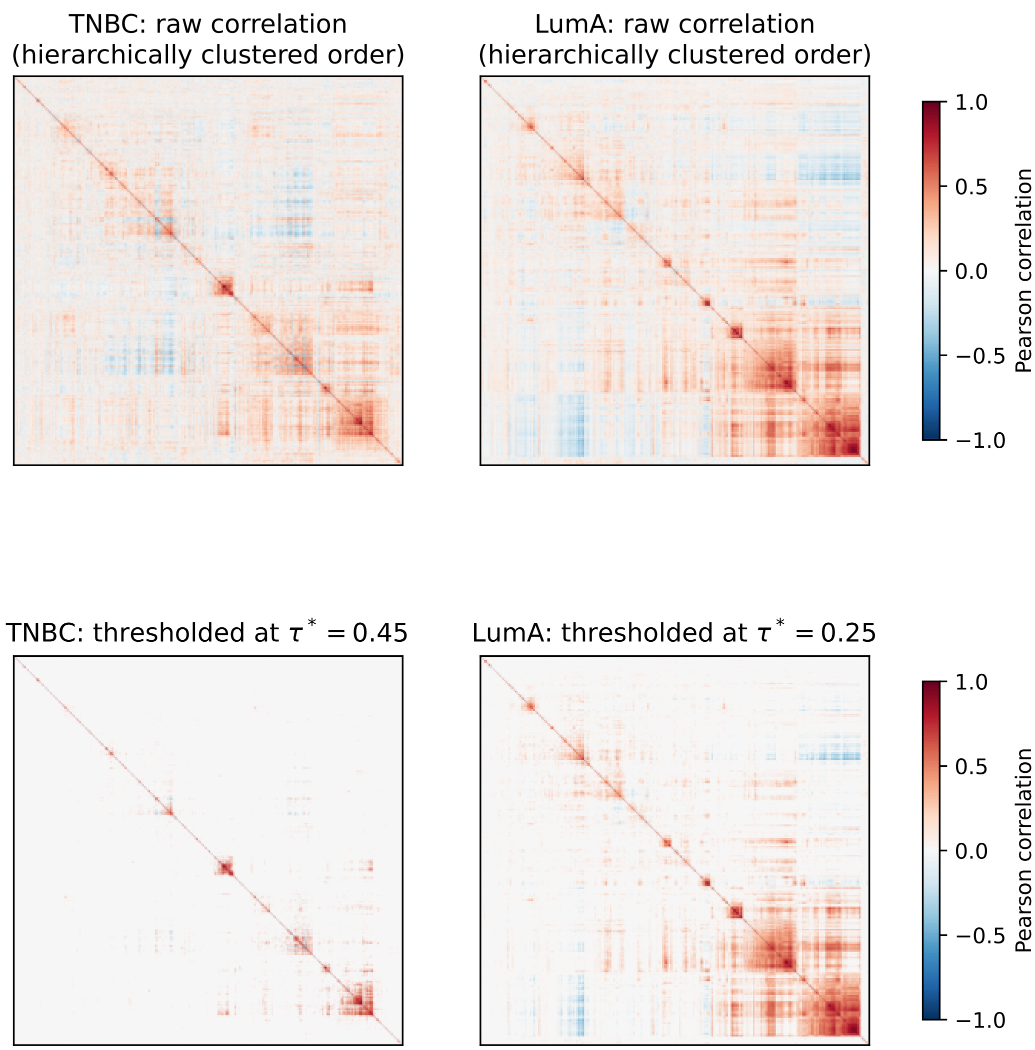

Figure 4: Raw and thresholded gene–gene correlation matrices for both cohorts, genes reordered by average-linkage hierarchical clustering on $1 - |C|$ so block structure is visible. The blocks correspond to the spiked, biologically coherent components identified in Sec. 2.2; LumA's visibly larger, denser red blocks match its lower τ^* and larger spiked-eigenvalue count. Generated by `scripts/11_heatmaps.py`.

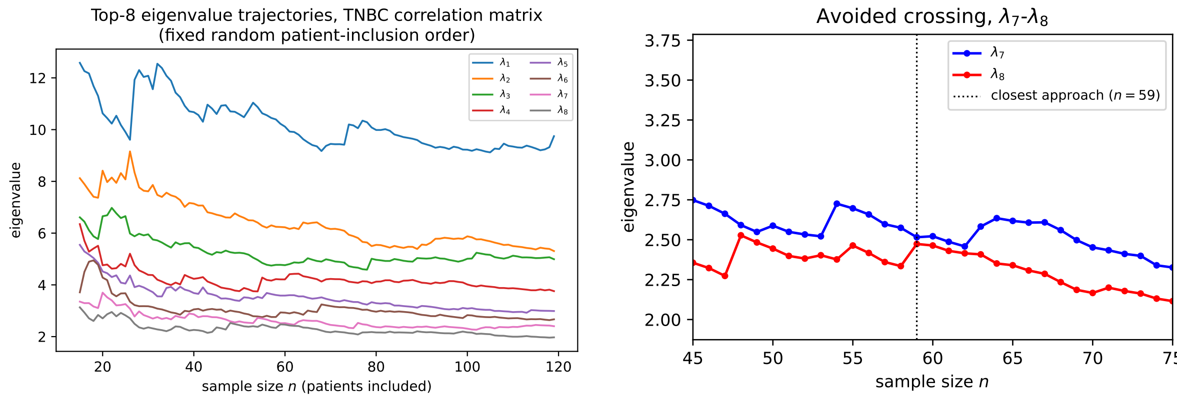

Figure 5: Left: top-8 eigenvalue trajectories of the TNBC correlation matrix (60 most variable genes) as sample size n grows from 15 to 119, patients added one at a time in a fixed random order. Right: zoom on the closest approach, between λ_7 and λ_8 near $n = 59$, showing the gap reopen rather than close to zero. Generated by `scripts/07_make_figures.py`.

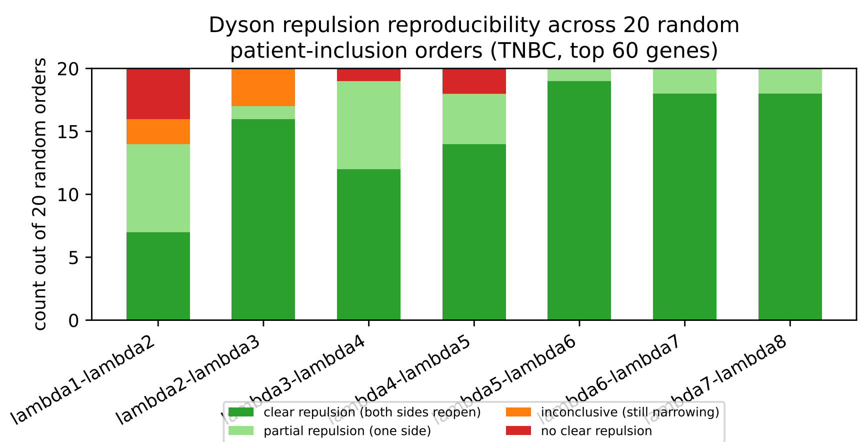

Figure 6: Reproducibility of the Dyson-repulsion pattern across 20 independent random patient-inclusion orders, TNBC, top 60 genes, for each adjacent eigenvalue pair. λ_2–λ_3, reported as inconclusive under the single order in Fig. 5, in fact shows clear repulsion in 16/20 orders; only λ_1–λ_2, involving the most dominant (spike) eigenvalue, is not reliably repulsive. Generated by `scripts/18_repulsion_reproducibility_figure.py`.

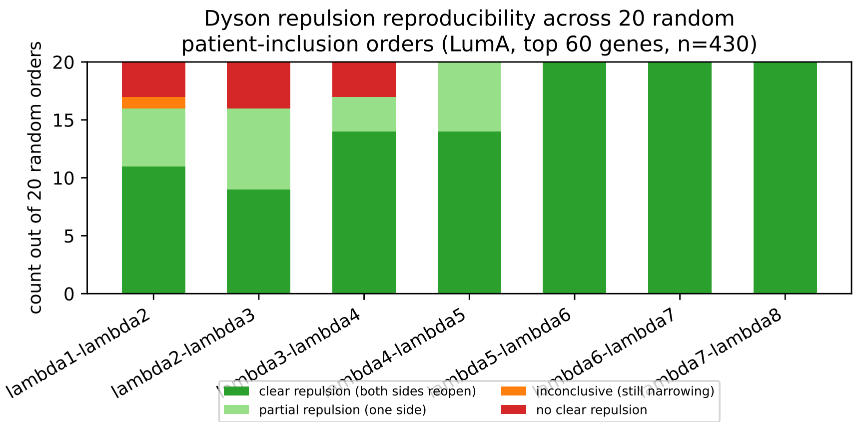

Figure 7: Dyson-repulsion reproducibility across 20 independent random patient-inclusion orders, LumA, top 60 genes, $n = 430$ — same method and criterion as Fig. 6 (TNBC). The two least-reliable pairs are again the ones adjacent to the largest eigenvalues. Generated by `scripts/27_luma_dyson_repulsion.py` and `scripts/29_new_figures_round3.py`.

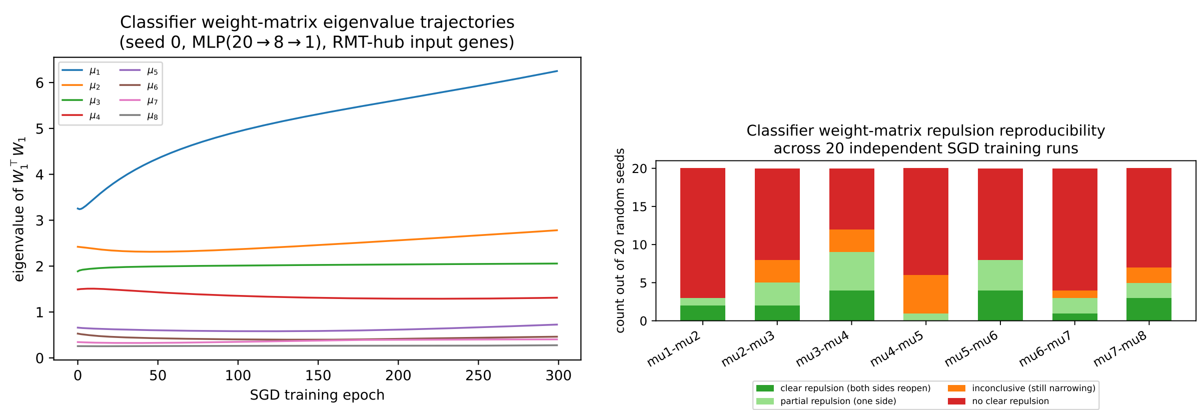

Figure 8: Left: $W_1^\top W_1$ eigenvalue trajectories across 300 SGD training epochs, one representative seed. Right: reproducibility of the repulsion pattern across 20 independent seeds, same classification as Fig. 6 — no pair matches the data side's reliability. Generated by `scripts/19_dyson_classifier_weights.py`, `scripts/20_dyson_classifier_weights_multi_seed.py`, and `scripts/21_classifier_dyson_figures.py`.

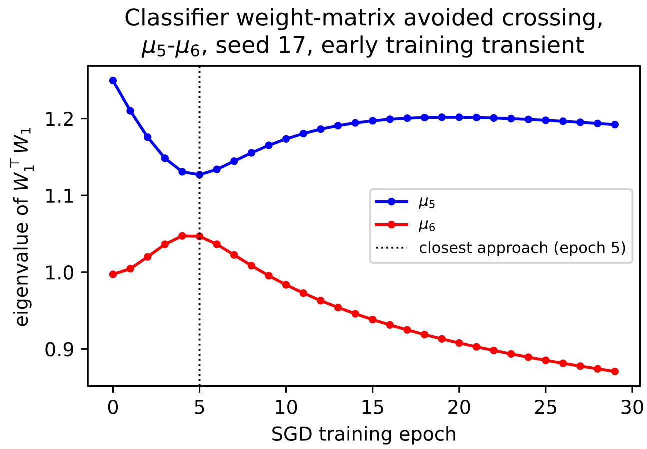

Figure 9: A clean example of the early-transient repulsion signature: μ_5-μ_6, seed 17, closest approach at epoch 5, gap reopening by epoch 29 — the same approach-then-reopen shape as Fig. 5 on the data side, here confined to the pre-convergence window (Sec. 2.11). Generated by `scripts/21_classifier_dyson_figures.py`.

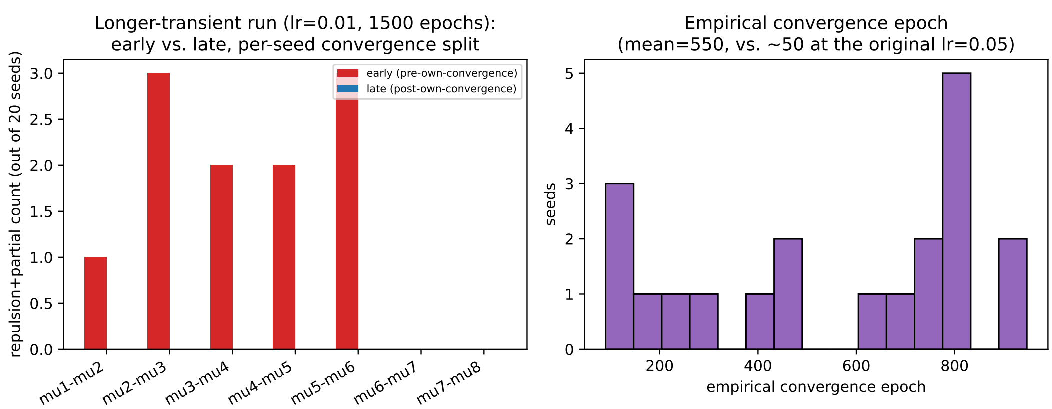

Figure 10: Left: early (pre-own-convergence) vs. late (post-own-convergence) repulsion-or-partial counts, by pair, at 5× smaller learning rate / 5× more epochs, each seed split at its own empirical convergence point. Right: distribution of empirical convergence epochs across the 20 seeds (mean 549.8, vs. $\sim$ 50 at the original learning rate) — convergence time lengthens by roughly 11× for a 5× smaller learning rate, not a proportional 5×. Generated by `scripts/28_classifier_dyson_longer_transient.py` and `scripts/29_new_figures_round3.py`.

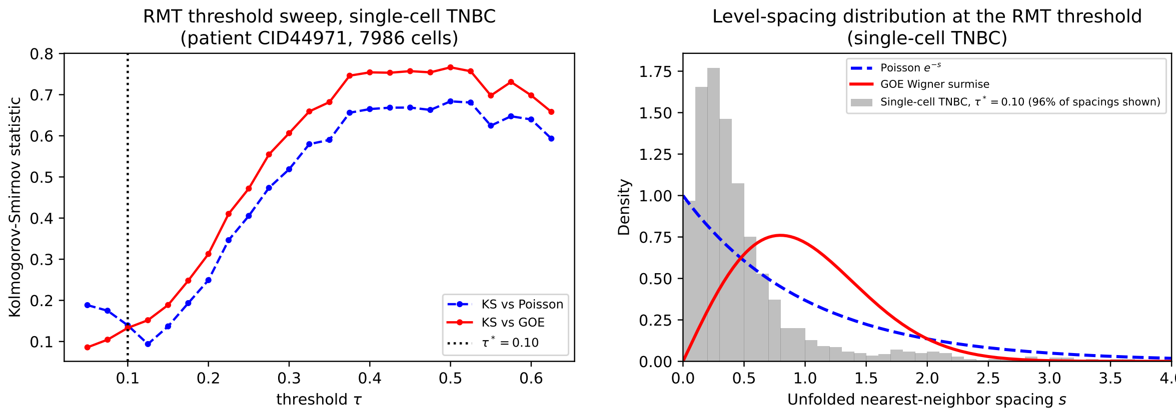

Figure 11: Left: RMT threshold sweep for the single-cell TNBC correlation matrix, same method as Fig. 3. Right: unfolded nearest-neighbor spacing distribution at $\tau^* = 0.10$, against the Poisson and GOE null curves, same method as Fig. 3 (4% of spacings fall above $s = 4$ and are excluded from the histogram, see text). Generated by `scripts/23_singlecell_line4_rmt.py` and `scripts/25_singlecell_figures.py`.

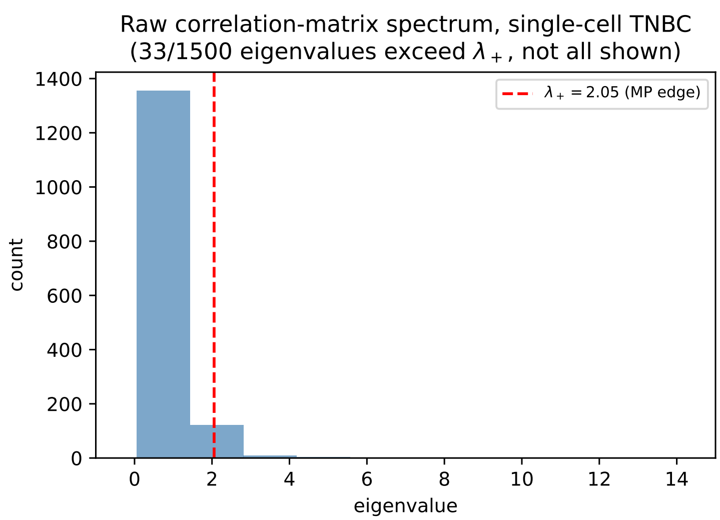

Figure 12: Raw correlation-matrix eigenvalue spectrum, single-cell TNBC, against the Marchenko–Pastur bulk edge $\lambda_+ = 2.055$ (Table 8). Generated by `scripts/23_singlecell_line4_rmt.py` and `scripts/25_singlecell_figures.py`.

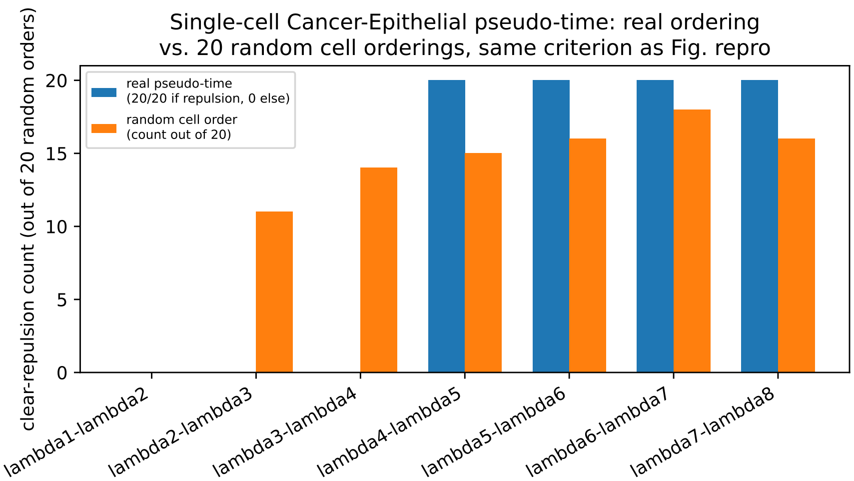

Figure 13: Single-cell Cancer-Epithelial pseudo-time Dyson-repulsion scan: clear-repulsion rate under the real pseudo-time ordering (blue, all-or- nothing since there is only one such ordering) vs. under 20 independent random cell orderings (orange, count out of 20). Random orderings match or exceed the real ordering for every pair with a clear signal, consistent with repulsion here reflecting growing sample size rather than this specific ordering's biological content. Generated by `scripts/26_singlecell_epithelial_pseudotime.py` and `scripts/29_new_figures_round3.py`.

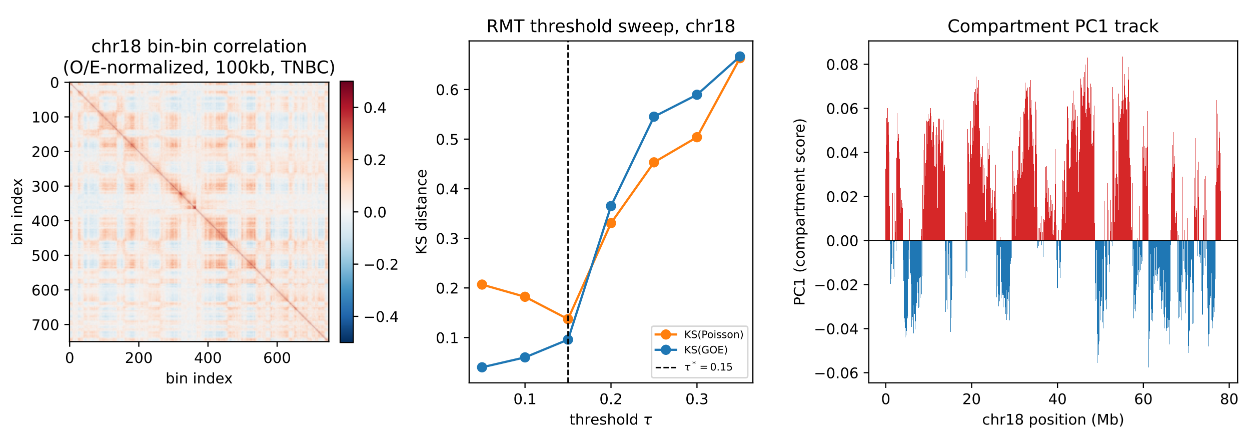

Figure 14: Real TNBC Hi-C, chr18, 100kb resolution (GSM5098082, 31). Left: O/E-normalized bin-bin correlation matrix, showing the plaid pattern characteristic of A/B chromatin compartments [21]. Center: RMT threshold sweep, same method as Fig. 3, selecting $\tau^* = 0.15$. Right: the top eigenvector's sign pattern along the chromosome, forming contiguous blocks, not scattered signs. Generated by `scripts/30_hic_line3_rmt.py` and `scripts/31_hic_figures.py`.

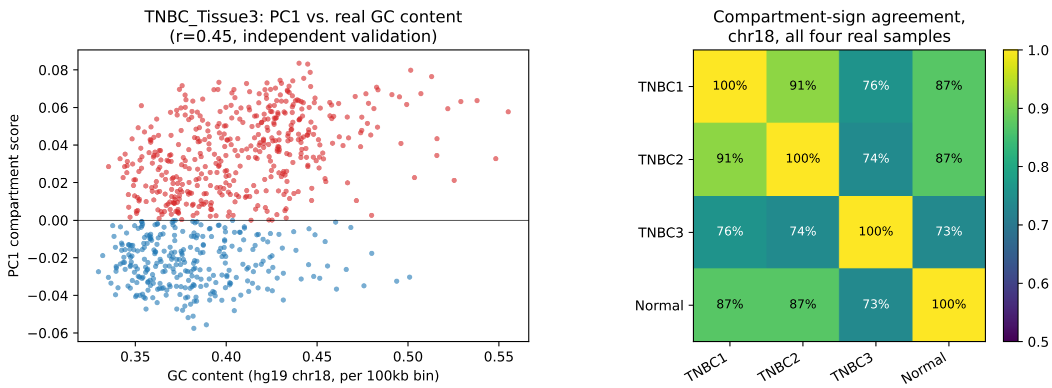

Figure 15: Left: TNBC_Tissue3's compartment PC1 score vs. real GC content per bin (hg19 chr18), colored by compartment sign — the independent validation the single-sample result needed. Right: pairwise compartment-sign agreement across all four real GSE167150 samples; TNBC_Tissue3 is the outlier against the other three, not the two tumors clustering against the one normal sample. Generated by `scripts/32_hic_compartment_validation.py`, `scripts/33_hic_multi_sample.py`, and `scripts/35_hic_validation_figures.py`.